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The Math & the Territory: On the Scientific Uses & Abuses of Machine Learning & Other Mathematical Modeling Strategies

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By

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ABSTRACT

This dissertation addresses, from the vantage point of philosophy of science, the human use and misuse of mathematical and computational tools, in particular, machine learning (ML) based methods. I introduce novel conceptual machinery for clearing up a number of debates over the application of formal methods in science which have stymied scientific integrity and progress. These concern the nature and utility of a ML-based mathematical modeling framework in neuroscience called the free energy principle, the role of ML in science generally, the extent to which ML-based tools can be appropriately regarded as "theory-free", deficiencies and false dichotomies within traditional philosophical conceptions of mathematical or scientific representation, and an inferential fallacy that arises in scientific modeling: conceptual reification.

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INTRODUCTION

0.1 Overview

This dissertation addresses, from the vantage point of philosophy of science, the human use and misuse of mathematical and computational tools, in particular, machine learning (ML) based methods. I introduce novel conceptual machinery for clearing up a number of debates over the application of formal methods in science which have stymied scientific integrity and progress. These concern the nature and utility of a ML-based mathematical modeling framework in neuroscience called the free energy principle, the role of ML in science generally, the extent to which ML-based tools can be appropriately regarded as "theory-free", deficiencies and false dichotomies within traditional philosophical conceptions of mathematical or scientific representation, and an inferential fallacy that arises in scientific modeling: conceptual reification.

0.2 Chapter 1. Navigating the Free Energy Principle

The first chapter of my dissertation examines a machine learning-based modeling framework in neuroscience, the free energy principle (FEP). The FEP is a mathematical framework developed by a prominent neuroscientist which draws formal motifs heavily from machine learning and various models in physics—including statistical mechanics, fluid dynamics and, increasingly, quantum mechanics Friston et al. (2006b); Friston (2009, 2010, 2012b, 2019). The framework is typically leveraged to model biological systems (brains, biomes, swarms, cells, tissues, individual organisms) as engaged in statistical inference over an external environment. How to evaluate the FEP—both in terms of its conceptual content and its epistemic status—has remained a subject of contention since its initial introduction into the literature, and has frustrated attempts to put it to use in scientific practice.

In the course of their work, scientists employ a variety of conceptual tools, including theories, models, laws, principles, formalism, thought experiments, and the like. The conceptual tools of science differ along a number of dimensions, including along axes of empirical content, truth value, and representational status. In its relatively brief tenancy in the literature, the FEP has been referred to promiscuously as nearly every art of conceptual instrument. Colombo and Wright (2018) observe that the FEP has been referred to as "a postulate, an unfalsifiable principle, a natural law, and an imperative" (Colombo and Wright, 2021, 3463). Raja et al. (2021) note that the terms of art used to refer to the FEP have included "a unified brain theory...a general principle for the self-organization of biological systems, and...a principle for a theory of every thing" (Raja et al., 2021, 49).

Such terminological confusion has been matched only in methodological confusion: while the literature on the FEP continues to expand, the principle has evaded incorporation into productive scientific workflow. Cognitive and neuroscientists have pinned its inability to get any empirical traction on a lack of falsifiability (e.g., Gershman (2019b)). Alongside almost universal disagreement over what sort of conceptual instrument the FEP is and its usefulness to science, there exists also an abundance of heavily conflicting accounts of the conceptual content of the FEP. Some have claimed that the FEP vindicates Freudian psychoanalytic theories while others claim that it reveals the secrets to tissue patterning in embryonic development. It has been held up at once as both embodied, enactive theory of cognition and as model of the Cartesian theatre.

In this chapter, I propose that a conceptual distinction be drawn between the representational vehicle—mathematical modelling framework—of the FEP and the conceptual or empirical content it is accorded when it is interpreted as a representation of some particular class of natural system. References to "the FEP" can thus be sorted into claims about the mathematical infrastructure of the general modelling framework, and specific models instantiated utilising the framework.

This philosophical intervention on the FEP literature makes clear the conceptual commitments of the framework and its utility to science for practitioners seeking to utilise the FEP in a number of ways. First, it makes apparent that disputes over the genuine commitments of the framework are profitless: the commitments are supplied to the framework when instantiated into a model, and there are nearly as many of these as there are papers written about the FEP. Second, it makes clear that claims about the falsifiability of the framework are missing their targets. The framework as a whole is not a candidate for empirical substantiation or falsification, but rather individual models are. Models built under the FEP are, in general, however, not built for the purpose of expressing contingent truths about nature which can be subjected to empirical test. They serve instead as a kind of formal thought experiment: a way of honing our conceptual schema as they relate to phenomena in nature prior to any experimental acquaintance with these phenomena.

0.3 Chapter 2. ML in Science: A Theory-Laden Conception

The second chapter of my dissertation evaluates a claim levied by various philosophers of science about the *atheoreticity* of machine learning methods when compared to the scientific use of classical mathematical or statistical tools. Machine learning (ML) refers to a class of computer-facilitated methods of statistical modelling in which patterns are inferred from data. As the use of ML-based technologies becomes ubiquitous across the sciences and across society, "AI hype" narratives engender misuse by misrepresenting ML's core epistemic capabilities.

One prominent and pernicious example is the presentation of ML as enabling "atheoretical" inference. By this, it is meant that the methods do not rest essentially on prior conceptualisations of the nature of the target phenomena. Touted as an epistemic virtue by some and decried as an impediment to epistemic aims by others, the theory-free conception of ML can be seen implicit in the uncritical ways in which ML tools are typically deployed. This false narrative is unfortunately lent credence by philosophers of science seeking to differentiate ML from traditional mathematical and computational methods.

Representatives from across the sciences, engineering, and philosophy of science have argued that ML will radically disrupt scientific practice or the variety of epistemic outputs science is capable of producing—call this the *disruption claim*. The disruption claim in turn rests on a *distinctness claim*. Proponents of the disruption claim hold this view because they take ML to exist on novel epistemic footing relative to classical modelling approaches utilised in science; namely, they take modelling with ML to be a "theory-free" enterprise in a way that renders it distinct from existing scientific methods.

I deliver an exposition of the operation of ML systems in scientific practice and reveal it to be a necessarily theory-laden exercise. Theory-ladenness comes in at three critical junctures: data provenance and processing, model architecture and training, and evaluation and use. The methods of ML, like existing scientific modelling techniques, are theory-laden in the sense of resting essentially on input from human conceptual grasp on the target phenomena, domain expertise, and established empirical knowledge. There is, therefore, no reason to believe that ML differs fundamentally in its reliance on theory from established mathematical modelling approaches in the sciences. This undercuts claims of epistemic distinctness and therefore at least one route to claims that ML will "revolutionise" scientific methods or outputs.

It is important to disabuse the scholarship of these distinctness and disruption claims about the operation of ML-based tools in science because they are facilitative of poor scientific practice with ML. Rather than freeing us from the constraints of preexisting conceptualisations of phenomena, I demonstrate that the theory-free ideal only serves to facilitate such unsubstantiated beliefs slipping in the back door of the inference procedure. Reification of this variety plagues the use of ML-based methods in both scientific and socially-sensitive contexts. Eliminating these conceptual errors relies on appropriately characterising the epistemic status of ML methods.

0.4 Chapter 3. Models of Data & Models of Theory: A Distinction without a Difference?

Among the primary subjects of philosophy of science are the conceptual instruments of science: theories, models, conjectures, and equations. How should we define these conceptual objects? How do they relate to one another? How do they relate to the world? Most importantly, how do they facilitate the acquisition of knowledge about that world? For as long as philosophers have regarded the scientific model to be an essential tool in science, they have conceptualized of models as falling into two broad categories: models of theory, and models of data. The account of what these are, and what differentiates them, was developed under a paradigm in philosophy of science known as the "semantic view of theories." The semantic view remained the dominant position in the philosophy of science from its introduction in the 1960s until the 2000s, during which time it was quietly unseated in popularity by a 'pragmatic' turn. Throughout this transition, the distinction between models of data and models of theory has persisted, playing important roles in accounts by philosophers identified with the pragmatic paradigm. Much—if not all—of what the original distinction consists in, however, relies on the ontological and epistemological commitments of the semantic view. In Chapter 3, I urge the case that the original distinction between theoretical and data models cannot be maintained. If philosophers of science within the pragmatic tradition are to deem such a distinction necessary or useful, it will need to be created anew on different grounds. I suggest that the most plausible reason to desire such a distinction is its apparent ability to differentiate between models whose empirical content is conjectural and models whose empirical content is derived from measurement or observation. The original distinction between theoretical and data models, however, only partially tracks the line between conjectural and empirically validated content. The most promising contender for a new taxonomy of models is, I urge, one grounded in the degree of confirmation of a model's empirical content.

0.5 Chapter 4. Scientific Representation & Conceptual Reification

This chapter characterized the fallacy of conceptual reification, critically examining accounts of scientific representation and how they interface with the characterization of this error. In the recent history of philosophy of science, the nature of scientific representation has been one of the most generative—and most hotly contested—topics in the field. Perhaps surprisingly, philosophers have paid little attention to how scientific representation goes wrong in practice. I refer to the patterns of mistakes that emerge when the interpretation of models misses its mark as instances of *conceptual reification*. Reification occurs when, in the course of interpreting a scientific model, some artifact of the representation is mis-mapped onto a target system, epistemically compromising the empirical inference at hand. I argue that specific features of a philosophical account of representation—namely, whether or not it conceives of representation as mind-dependent—impact our ability to handle reification. "Mapping accounts" of representation, on which representation is understood as a two-place, objective relation, obscure sources of reification. Agent-relative depictions of representation, meanwhile, facilitate our ability to identify, analyze, and correct for reification. Insomuch as commitments to the nature of representation guide scientific use of models, triadic accounts help modelers to avoid reification.

CHAPTER 1

NAVIGATING THE FREE ENERGY PRINCIPLE

1.1 Introduction

Much has been written about the free energy principle (FEP), and much misunderstood. The principle has traditionally been put forth as a theory of brain function or biological self-organisation. Critiques of the framework have focused on its lack of empirical support and a failure to generate concrete, falsifiable predictions. I take both positive and negative evaluations of the FEP thus far to have been largely in error, and appeal to a robust literature on scientific modeling to rectify the situation. A prominent account of scientific modeling distinguishes between model structure and model construal. I propose that the FEP be reserved to designate a model structure, to which philosophers and scientists add various construals, leading to a plethora of models based on the formal structure of the FEP. An entailment of this position is that demands placed on the FEP that it be falsifiable or that it conform to some degree of biological realism rest on a category error. To this end, I deliver first an account of the phenomenon of model transfer and the breakdown between model structure and model construal. In the second section, I offer an overview of the formal elements of the framework, tracing their history of model transfer and illustrating how the formalism comes apart from any interpretation thereof. Next, I evaluate existing comprehensive critical assessments of the FEP, and hypothesize as to potential sources of existing confusions in the literature. In the final section, I distinguish between what I hold to be the FEP—taken to be a modeling language or modeling framework—and what I term "FEP models."

The questions most frequently—and most fervently—asked about the FEP are: Is it true? What is it true of? How do we know (empirically) that it is true? These questions, I argue, rest on a category mistake. They presume that the FEP is the sort of thing that makes assertions about how things are, cuts at natural joints, and can be empirically verified or falsified. I urge that before we can make serious headway on understanding the FEP and putting it to work in scientific practice, we must answer to an entirely different set of questions: What sort of scientific object is the FEP? To what discipline(s) does the FEP belong? What role is it intended to play in relation to empirical research? Does the FEP even properly belong to the domain of science? The extant literature has largely begged, dodged, dismissed, and skirted around these questions, without ever addressing them head-on. To the extent that existing works have attempted to address these questions, all have proceeded under the—mistaken, or so I will argue—assumption that the FEP is, itself, fundamentally truth-apt. These questions must, I urge, be answered satisfactorily before we can make any headway on the theoretical consequences of the FEP. Empirical work with the FEP has proceeded in the absence of such a clarificatory project, but it has not been unencumbered by it. I take preliminary steps towards answering these questions in this chapter, first by examining the historical path traversed by key formal elements of the framework and the implications they hold for its utility, and second, by offering a route to interpreting the FEP, and models built therefrom, in light of an abundant philosophical literature on scientific modeling.

Existing literature on the FEP invokes "the free energy principle" to refer indiscriminately to both the raw formal structure of the framework and to various models constructed therefrom. My novel proposal is that we reserve the term "free energy principle" to designate the model structure, which can be differentiated from distinct models composed from the combination of that structure with a scientist's or theorist's construal thereof. To this end, I first deliver an account of what is known as model transfer, which illustrates a phenomenon undergone by the FEP and helps us to see how models can be broken down to a structure and a construal. I also broach the topic of conceptual reification in modeling. Next, I trace out the history of the core formal elements of the FEP, illustrating the formal skeleton of the FEP, sans interpretation. In section three, I tackle the claims made in existing critical assessments of the FEP, elucidating where these have gone wrong in their interpretation of the framework. From there, I look to the literature on scientific modeling once again, drawing conclusions about how to wield and interpret the various scientific models built from this formal foundation. My hope is that this text can serve as a fruitful starting place for philosophers and scientists looking to utilize the FEP formalism.

1.2 Model Transfer, Structure & Construal

This section introduces the notions of model transfer, model structure, model construal, and model reification, which will enable us to better understand the FEP, along with its uses and misuses. According to several popular accounts of scientific modeling, a scientific model is composed of a structure and an interpretation. This breakdown is most applicable to abstract, formal models. It is perhaps easiest to see this distinction play out in cases where a model structure, originally designed for one modeling purpose, is exapted away from its original interpretation and lent a new one.

1.2.1 The Lotka-Volterra Model

Take the Lotka-Volterra model. The structure, in this case, is a system of nonlinear differential equations. The Lotka-Volterra model originated in physics and chemistry. The equations were originally proposed by Alfred Lotka in 1910 as a model of autocatalysis—self-catalysing chemical sets (Lotka, 1910). They describe the rate of change of chemical concentrations as a chemical system pushed out of equilibrium restores itself, by oscillations, back to steady state. A linearization of the model produces a system similar to a harmonic oscillator.

A paradoxical trend was noted in fish populations of the Adriatic Sea surrounding World War I. Italian biologist Umberto d'Ancona measured the relative prevalence of fish of various species. He found that, during WWI, when fishing in the Adriatic Sea all but ceased, the predator population experienced a boom. In response, the prey populations diminished considerably. When fishing resumed after the war—an indiscriminate biocide—the prey population soared. Vito Volterra (1926) employed a system of nonlinear differential equations formally equivalent to Lotka's (1910) chemical model to explain how the removal of a constraint (fishing) increased the amplitude of predator population size, while the reimposition of this constraint—the return of fishing after the war—increased the amplitude of the prey populations. In 1956, Lotka independently proposed the same set of equations for the purpose of explaining predator-prey dynamics in his *Elements of Mathematical Biology* (Lotka, 1956).

How does this exemplar help us to understand the free energy principle? In two ways. First, it is common for a relatively coarse-grained formal model initially utilized in one domain (in this case, physical chemistry) to be later imported into an altogether different discipline (in this case, population biology, ecology, and ethology). Second, the case renders intuitive the distinction between the structure and the construal of a model. The structure—a system of differential equations—remains the same for both the (1910) model of catalytic sets and the (1926, 1956) models of predator-prey dynamics. The two models differ in that, in the first instance, scientists interpret the equations to represent chemical concentrations, while in the second instance, scientists interpret the equations to represent population density.

Conceptual reification is a common ailment of scientific modeling. It is particularly likely to occur in cases in which models have somewhat convoluted histories. Reification involves mistaking an aspect of a model—its structure, its construal, or the union of both—for an aspect of empirical data or the natural world; mistaking the math for the territory, so to speak. Reification also occurs when we take an analogical relationship to be a literal one, or when elements of a model's construal in its original domain of application get brought along parasitically into a novel domain in model transfer.

In the next section, we will examine the historical trajectory undertaken by several of the key formal elements of the framework; its legacy of model transfer. This historical exercise allows us to clearly separate the formalism from its interpretations in various domains and lends us insight into some of the reification that has led to confusions surrounding the FEP.

1.3 The Free Energy Principle

1.3.1 History of the Formalism

To understand the FEP, separated from the various construals attached to it in its various contemporary instantiations for theoretical or simulation purposes, and separated from various construals which may have attached themselves parasitically to the framework during its elaborate history of model transfer, it will be necessary to trace out this historical record. Many of the formal tricks embodied in the FEP originated in physics, some in machine learning, some in physics via machine learning. The FEP, however, is not a law of physics, nor is it a theory, model, or principle belonging to the physical domain. It is not a machine learning technology, though it draws upon some of the same underlying statistical techniques.

A relevant contingent of people concerned with the FEP take it to be, in one way or another, a *physical* description of natural systems. This has an obvious form: taking notions such as energy, entropy, dissipation, equilibrium, heat, or steady state, which play important roles in the FEP, in their senses as originally developed in physics. There is a more subtle form of this tendency, however, in which people begin with the assumption of an analogical relationship to physics, or a mere formal equivalence, but conclude that the formalism of the FEP nonetheless picks out real and measurable properties of natural systems, albeit perhaps more loosely and abstractly than its physical equivalents would. This, I argue, is a conceptual reification; a vestigial interpretation from the formal methods' origin in statistical physics.

1.3.2 The Epistemic Turn in Statistical Mechanics

An important precursor to the FEP that seldom comes up in the literature is Jaynes' maximum entropy principle.[1] The classical interpretation of statistical mechanics views the macroscopic variables of some physical system of interest—say, heat, volume, and pressure—as physical constraints on the microscopic behaviour of the system. This is a decidedly physical interpretation of the maths. The critical insight in Jaynes (1957) was that we could give this all a subjectivist, epistemological reading, casting these macroscopic variables as knowledge about the system, with the lower-order details to be inferred. The principle of maximum entropy guarantees the maximum (information) entropy of a probability distribution given known variables. Maximizing the entropy of the distribution guarantees that we are not building in any more assumptions than we have evidence for. This principle of maximum entropy took the formalism of statistical mechanics and gave it an information-theoretic interpretation, turning the second law of thermodynamics into a sort of Occam's razor for Bayesian inference. This is because the maximum entropy principle brings us to adopt the probability density with the most widely dispersed probability density function, given the known variables, just as entropy will be maximized with respect to macroscopic variables in statistical mechanics. These are formally identical. Given the frequency with which the literature on the FEP makes reference to Jaynes, one might think it a rather inconsequential piece of the puzzle. In order to understand the FEP, however—and why it is closer to a statistical technique than it is to a falsifiable theory of biological self-organisation—it is important to see that there is a clear precedent for leveraging the maths of statistical mechanics as a method for Bayesian inference. Jaynes' maximum entropy principle (often referred to as MaxEnt) has had tremendous success as a tool for scientific modeling across the sciences. The free energy principle, much like the maximum entropy principle, takes the mathematical machinery of statistical mechanics and lends the formal tools therein a

1. For a thorough overview of the FEP/MaxEnt connection, refer to Gottwald and Braun (2020).

distinctly epistemic, inferentialist bent.

1.3.3 The Mean Field Approximation

Independently, an approach known as mean field theory emerged in statistical mechanics at the beginning of the twentieth century that enabled physicists to study high-dimensional, stochastic systems by means of an idealized model of the system that would average out, rather than summing over, the interactions of elements within the system. Feynman (1972) introduced what are known as variational methods within the path-integral formulation of mean field theory. By exploitation of the Gibbs-Bogoliubov-Feynman inequality, one is able to achieve a highly accurate approximation of the energetics of a target system under a range of conditions. This is accomplished via minimisation of free energy by variations on a simplified candidate Hamiltonian to bring it into accord with the true Hamiltonian.[2] What is important to understand about Feynman's original formulation of the free energy minimisation technique is that it is 1. not to be taken as a literal representation of a target system but rather it is a formal trick for approximating otherwise intractable computational problems that arise in dealing with certain physical systems, and 2. that the free energy involved nonetheless refers to a physical quantity: Helmholtz free energy.

1.3.4 Free Energy in Machine Learning

The method of variational free energy minimisation was adapted for statistics and machine learning towards the end of the twentieth century as *ensemble learning* or *learning with noisy weights* (Hinton and Van Camp, 1993; Hinton and Zemel, 1993; MacKay, 2001). Thus free energy minimisation in statistics is a variational method for approximate inference where intractable integrals are involved. A quantity, termed variational free energy, is minimized,

2. We may think of the Hamiltonian of a physical system as the net kinetic and potential energies of all of the particles in the system.

thus bringing the ensemble density or variational density—the approximate posterior probability density, on a Bayesian interpretation—into approximate conformity with the true target density (Friston et al., 2006a; Hinton and Van Camp, 1993; MacKay, 1995a,b,c; Neal and Hinton, 1998). According to Friston, this method of approximating the posterior density or ensemble density is a statistical analogue of the mean field approximation in statistical physics (Friston et al., 2006a). We can see that both the free energy term and the construct it is being leveraged to approximate refer to energetic properties of the physical systems under study–Helmholtz free energy, and the system's Hamiltonian—as the method was originally purposed by Feynman (Feynman, 1972). The variational free energy and the variational or posterior probability density involved in the variational free energy minimisation technique as employed by Hinton and Van Camp (1993), however, are purely statistical constructs. Thus, although it may be tempting to lend the "free energy" under the FEP a physical interpretation, it is not meant to invoke a physical quantity. Variational free energy is not Helmholtz free energy, despite the formal similarity.

1.3.5 Variational Bayes

The finer points of the formulation of variational Bayes in use today were worked out by Beal (2003) and Attias (2000). Beal (2003) illustrates how conceiving of approximate Bayesian inference in terms of conditional probabilities can be facilitated via graphical models, such as Markov random networks, highlighting the importance of the set of nodes that form the Markov blanket of the set of interest. An exact deployment of Bayes' theorem almost always leads to intractable integrals—the sort of calculus it would take an adept mathematician years to solve. By contrast, approximate variational Bayesian methods generate candidate probability distributions and assess the Kullback-Leibler (K-L) divergence between candidate and target distributions.

1.3.6 Innovations in Friston's Free Energy Minimisation

Karl Friston took the method of variational free energy minimisation and gave it a dynamical-systems interpretation, specifying the free energy minimisation dynamic in terms of the Fokker-Planck equation and, in particular, the solenoidal and irrotational flows that fall out of the Helmholtz decomposition thereof, of which the irrotational flow can be conceptualised as a gradient ascent on an attracting set (Friston, 2009, 2010, 2012a, 2019; Friston and Stephan, 2007; Friston et al., 2008). This allows us to think of free energy minimisation simultaneously as a method of approximate Bayesian inference and as a flow.[3]

Jaynes' maximum entropy principle took the formalism of statistical mechanics and leveraged it to accommodate the process of inference given limited and noisy data. Friston's FEP goes a step further, borrowing mathematical tools from the physical sciences to enable the formal representation of processes of inference about the (stipulated) inferential dynamics of systems in nature.

1.3.7 Fundamentals of the FEP

The Fokker-Planck, or Kolmogorov Forward equation describes the time evolution of a probability density function. The Fokker-Planck equation originated in statistical mechanics, in which it described the evolution of the probability density function of the velocity of some particle, or its position, in which case it was known as the Smoluchowski equation. In the context of the FEP, the Fokker-Planck equation describes the evolution of the probability density function of the state of a system. As such it can be thought of as a trajectory through one abstract state space which is a probabilistic representation of some lower-order abstract state space representing what state a given system is in over some definite time window.

3. Friston (2009) notes that it is interesting that the formulation of free energy minimisation using gradient flows (otherwise known as gradient descent) was an important practical development in the data analysis tools commonly applied in neuroscience—for example, in dynamic causal modeling. In brief, this freed one from the analytic derivations of vanilla variational Bayes and the use of conjugate priors; enabling a generic variational scheme for modeling empirical data known as variational Laplace.

Any vector field that satisfies the appropriate conditions for smoothness and decay can be broken down into solenoidal (curl) and irrotational (divergence) components. This is known as the Helmholtz decomposition; the fact that we can perform the Helmholtz decomposition is then known as the fundamental theorem of vector calculus.

The static solution to the Fokker-Planck equation is a probability density termed the Nonequilibrium Steady State density, or NESS density (Friston, 2019); (Friston and Ao, 2012b). The notion of nonequilibrium steady state is native to statistical mechanics, wherein it describes a particular energetic dynamic between a system and its surrounding heatbath. NESS is best understood as the breaking of detailed balance. Detailed balance is a condition in which the temporal evolution of any variable is the same forwards as it is backwards (the system's dynamics are fully time-reversible). Detailed balance holds only at thermodynamic equilibrium. In nonequilibrium steady state, balance holds in that none of the variables that define the system will undergo change on average over time, but there is entropy production, and there are flows in and out of the system. (Jiang et al., 2004) and (Zhang et al., 2012) have demonstrated that nonequilibrium steady state can be represented as a stationary, irreversible Markov process. This development paved the way towards a purely statistical rendering of the notion of NESS.

The literature on the FEP also rests centrally on the notion of a Markov blanket (Kirchhoff et al., 2018), an adaptation of Pearl's (Pearl, 1988) Markov boundary. A Markov blanket essentially partitions the world into a thing which can be conceived of as, in its very existence and dynamics, performing a kind of inference, and a thing it is inferring—on a yet more basic level, the Markov blanket allows us to partition the world into a system of interest, and all that lies outside of that system of interest. Systems are represented under the FEP as being subject to random fluctuations, which are responsible for the stochasticity of the systems involved. These fluctuations would result in the dissolution of the systems of interest, were it not for some balancing flow. In the absence of a counteracting flow, the system, as defined by its Markov blanket, would cease to exist as such. If the set of states considered to be the system (internal states and their Markov blanket) are to resist this dissipative tendency, they must counteract it. This counteracting flow can be conceptualized in a number of ways. For one, we can think of the perturbations as causing the NESS density to disperse, and the irrotational flow under the Fokker-Planck equation as countering these fluctuations. We can also think of it as ascending the gradients induced by the logarithm of the NESS density. The system is hillclimbing on a landscape of probability. It seeks to ascend peaks of maximum likelihood and escape from improbable valleys. In fact, the FEP is a form of dynamic expectation maximisation, which is itself a maximum likelihood function (Friston et al., 2008). The flow of the system must also, moment by moment, minimize surprisal or self-information by gradient descent. Variational free energy constrains this activity by placing an upper bound on surprisal.

Under the FEP, a system of interest can be represented as being subject to random perturbations, which would induce dissipation were it not for some flow countering this dissipation. The Fokker-Planck equation encapsulates these random perturbations as w—the Wiener process, or Brownian motion. The curl-free (irrotational) dimension of the flow described by the Fokker-Planck under the Helmholtz decomposition will be seen to counter this flux, maintaining the integrity of the NESS density, which places high probability mass over the system's pullback, or random global attractor (Friston, 2019). All this means is that, statistically speaking, the system prefers this region of its phase space—the way a cat likes a laptop computer or a ball likes to roll downhill. The NESS density can also be cast as a generative model, as the highest probability region of the system's phase space will be a joint distribution over all of the system's variables. By generative model, we mean here a joint probability distribution over external and blanket states. For this reason, we can conceptualize the behaviours of the systems treated under the FEP as statistical models of the causes impinging upon them from their environments. This follows from the complete class theorem which, in its Bayesian, statistical generalisation, states that any decision procedure operating according to a loss or risk function in a finite sample space is, under certain assumptions, Bayes optimal with respect to some prior. By extension, then, any dynamical system that minimizes some loss or risk function according to some decision procedure is taken to be Bayes optimal under some generative model and priors.

This brings us back to the inferential interpretation of the dynamic described by the FEP. When we apply Bayes theorem to a problem of inference or belief updating, we want to maximize marginal likelihood. Marginal likelihood is the likelihood of some observation given our model; it is also termed Bayesian model evidence, or simply evidence. Surprise and evidence are inverse functions. When we minimize surprisal, we are maximizing model evidence. Thus, systems under the FEP are said to be 'self-evidencing' (Hohwy, 2016). Over time and on average, the minimisation of surprisal minimizes information entropy. This effectively prevents a system's states from dispersing in a statistical sense—it keeps the values of certain key variables within certain existential (that is, definitive of the system) bounds. In minimizing (an upper bound on) surprisal, we minimize (a lower bound on) model or marginal evidence, or simply evidence. This makes free energy minimisation equivalent to evidence lower bound optimisation (ELBO), an objective function that anyone with a background in machine learning or Bayesian statistics will find themselves familiar with.

Here we have traced the history of key formal elements of the FEP in Jaynes, Feynman, Hinton, Pearl, Beal, and Friston. Having a handle on this history is necessary in order to grasp the subtle turn away from statistical approximations of physical properties of physical systems to a pure, substrate-neutral method of statistical inference. When we speak of annealing a model in statistical mechanics, ratcheting the temperature of the system up and down in the hopes of bumping it out of local minima, this does not refer to an act of literally injecting energy into a physical system to increase the speed of particle motion. It is a statistical analogue of a physical process. Likewise, the energy and entropy of the FEP are formal analogues of concepts defined in thermodynamics and statistical mechanics with a long history of use in information theory, statistics, and machine learning, in which they have lost their correspondence to any measurable properties of physical systems.

1.4 Critical Appraisals

Mine is not the first attempt to get to the bottom of the FEP. There have been, to date, a number of attempts at comprehensive critical assessment of the FEP, including (Colombo and Wright, 2017), (Colombo, 2017), (Colombo and Wright, 2021), (van Es, 2020), (Gershman, 2019b), (Gershman, 2019a), (Klein, 2018), and (Sims, 2016). The nominal worries of these critical accounts include that the FEP lacks biophysical or cognitive realism, that it somehow contravenes experimental observation, that it is incapable of providing an all-encompassing account of brain function, or that it fails to make novel predictions (Colombo and Wright, 2017); (Colombo, 2017); (Colombo and Wright, 2021); (Gershman and Daw, 2012); (Gershman, 2019a); (Klein, 2018). The real worry, though, the worry that is only made explicit a few beers in to the spillover of the conference proceedings into some smoke-filled local tavern, the worry that is only put to words in the manuscripts that get rejected before they ever make it to print, is that the FEP is somehow empty; that it lacks all conceptual content. My claim is that the FEP is, indeed, empty in just this way, that that is not—or ought not to be—a secret, and that its contentlessness does not count as a mark against the framework.

Indeed, existing works in this genre all stack the solutions they arrive at against this conclusion. They all appear to beg the question. These critical accounts, much like the preponderance of positive accounts, fail to differentiate between the FEP as *formal model structure* and the various models built from or atop this structure, which themselves are often casually referred to as "the FEP" in the literature.

1.4.1 Colombo & Wright

Colombo and Wright (Colombo and Wright, 2021) do entertain the idea that the FEP is merely a formal modeling tool—indeed, they even diagnose the problem of reification in modeling, "the risk of conflating scientific models and their targets" (p. 12)—but they never take this hypothesis seriously. Colombo and Wright (Colombo and Wright, 2021) raise the matter as a serious question to be addressed: "should we understand FEP as a modeler's tool to characterize and predict adaptive behavior, or should it be understood as an objective feature of target systems?" (pp. 15-16). As they proceed, however, they merely "assume that the probabilities involved in FEP aren't simply modelers' tools" (p. 17). It is unclear on what grounds they justify this assumption, beyond the further (unjustified) assumption that the aim of the modeling exercise must be realism and that "Friston (2013) seems to interpret the probabilities involved in FEP as objective features of real-world systems" (Colombo and Wright, 2021). Ultimately, the hypothesis that the FEP is empirically contentless is swept aside: "what's intended cannot be that FEP is unfalsifiable because it fails to be truth-apt" (pp. 22-23).

1.4.2 Gershman, Klein, & Williams

Gershman (Gershman, 2019a)'s supposed critique of the framework does not actually address the FEP, but diverts its attention to process models, writing "we will be concerned with [the FEP's] credentials as a theory, and therefore we will pay particular attention to specific implementations (process models)" (p. 2). Sims (Sims, 2016) likewise conflates the FEP with associated corollaries and process theories. For example, he writes: "In its form as a theory of cognition, the application of this theory to explain mental phenomena draws heavily upon the notion of expected precision" (p. 970). Expected precision is a notion proper to predictive coding and various models that fall under the heading of predictive processing. In a similar vein, Sims writes that the "free energy principle makes certain non-trivial predictions about brain structure and function," listing among these predictions a greater preponderance of top-down (feedback) connections than bottom-up, feedforward connections, and the organisation of the cortical hierarchy, likewise the purview of hierarchical predictive coding or predictive processing models. Klein (Klein, 2018) argues that the FEP is either susceptible to the dark room problem or it is something like an idealized model. On this deflationary depiction of the FEP, it is taken as "a starting point from which one might develop explanations," and its success (or failure) as a scientific tool ultimately rests on "the empirical adequacy of detailed models which spring from it" (p. 2554). This, I think, is precisely the right mode of understanding FEP-based models. Thus, models built from the formal architecture of the FEP offer "a deliberate simplification, which buys scientific fruitfulness at the cost of literal truth" (p. 2554). Notably, this is quite close to a Wimsattian (Wimsatt, 1987) view of the epistemic virtues of modeling. Williams (2020) delivers a description of the FEP that is diametrically opposed to Sims' (Sims, 2016) depiction: "the FEP does not advance a causal hypothesis. Specifically, it provides no information about how self-organization is causally generated and sustained in the systems that it applies to" (p. 20).

It appears that we have the theoretical equivalent of binocular rivalry when it comes to depictions of the FEP's empirical commitments and epistemic status. How, then, do we resolve this conflicting vision? First, we ought to note that both Sims (2016) and Williams (2020), like most of their predecessors, take "the free energy principle" to refer to both the raw formalism and to various models predicated on that formalism which Friston and colleagues have described over the years. Additionally, Klein (2018) seemingly takes "the free energy principle" to refer to active inference and perceptual active inference, hierarchical predictive coding, and various predictive processing models, describing all as one and the same theory of cognition. The deflationary conclusion he reaches, however, maps onto the role played by various models constructed from the formal framework of the FEP: FEP-based models act as heavily-idealized, generic or targetless mathematical models. One core function such models serve in relation to scientific practice is as a stepping stone on the road to more fine-grained and empirically rich models. Williams (2020), on the other hand, adequately differentiates the FEP from various associated process models and adjacent theories within the Bayesian brain canon. The conclusions he reaches in regards to the FEP's explanatory scope and epistemological status fail to cohere to what the literature has, to date, *descriptively* used "the FEP" to denote. However, I contend that Williams (2020)'s assessments of the FEP are precisely on the mark as an appraisal of what the FEP ought, *normatively*, to refer to: namely, the formal structure, absent any interpretation relating it to a target.

1.4.3 Van Es

van Es (2020) is the first to draw clear connections between FEP models and the literature on scientific modeling. Van Es's (2020) argument proceeds as follows: 1. FEP models describe organism-environment interactions in terms of modeling. 2. It is unclear to what extent it is intended that an FEP model models organisms as though they were, themselves, engaged in a practice of modeling their environments, and the extent to which proponents of FEP models intend the models posited thereunder to be literal, in the world, and independent of scientists' modeling practice. 3. The existing literature utilizing FEP models to address cognition seems to rest, fundamentally, on a conflation between the two. 4. Kirchhoff and Robertson (2018) argue that the sense in which organisms' dynamics mirror their environments, under FEP models, is not representational in nature, but merely covariational. 5. There is consensus in the literature on scientific modeling that scientific models are representational in nature. 6. The sole nonrepresentational account of scientific modeling on offer appeals to social practice of science to ground the representational features of models. 7. Models as leveraged by organisms, under FEP models, cannot recourse to the social practice of science to ground the representational features of their models. 8. Therefore, models posited under FEP models fail to count as models. 9. We must, then, according to van Es, adopt an instrumentalist stance on models under the FEP.

I believe that van Es errs, however, in the presumption that the generative models embodied or entailed by organisms under various FEP models are models in just the same sense in which scientific models are models. For it is not the case that all scientific models are generative models in the formal, statistical sense. The class of scientific models is not co-extensive with the class of statistical models of joint probability distributions. Why, then, should we assume that the generative modeling stipulated under the FEP is precisely the same sort of modeling that scientists engage in, and subject to the same constraints?

While instrumentalism with respect to the mathematical constructs leveraged in FEP models is a strong position, I believe that the argumentative route van Es traverses to arrive at this position is a nonstarter. Van Es contends that the existing FEP literature erroneously conflates two senses of modeling. In fact, van Es' own paper appears to exhibit a kind of conflation between various senses of the term. There are, in the first place, models as utilized by scientists to gain leverage over the natural world. There is a deflationary, statistical notion of a generative model as a statistical model of a joint probability distribution. Van Es (2020) seems to run the two together, and further conflates both with a third notion of models, under which a brain may be said to "model" its environment in some cognitivist, representationalist sense.

1.4.4 A Tentative Diagnosis

Whence such confusion? For one, we have seen the convoluted history of model transfer the FEP has undergone, with formal elements drawn from a number of disciplines and passing through multiple interpretations before achieving their current form and use. For another, it is not often pragmatically necessary in the practice or research or theorizing to specify in painstaking detail what formal models we are drawing from and in what mixture and quantity. For another, at the risk of psychologizing, when one's primary mode of relating to the world is not linguistic, but mathematical, the distinction between "maths" and "territory" makes little sense. In fact, I believe that a necessary precondition to being a good physicist is the loss of this distinction between literal description of the world and mere formal trick. This creates something of a barrier to interdisciplinary communication. Physicists or those with physics backgrounds are often known to say things which, to mathematical modellers in biology, cognitive science, pure maths, or machine learning often seem to mix metaphors. I view as symptomatic of this tendency Friston's overly literal descriptions of the formalism.

There are many places throughout the literature on the FEP in which the language used to describe the formalism can easily give rise to the misconception that the framework is a literal—perhaps physical—description of some measurable feature of natural systems, or cuts at natural joints. Ramstead et al. (2018), for example, write that "systems are alive if, and only if, there[sic] active inference entails a generative model" (p.33). Under the perspective of the FEP—that is, once we have elected to model biological systems using the formal tools the FEP provides us—any system we choose to model in this way will behave as the model dictates it must. Under the FEP, in order to be a system, certain mathematical assumptions must hold. In particular, we assume a weakly-mixing random dynamical system, a Markov blanket, and either ergodicity or (organisation to) nonequilibrium steady state (NESS). If we take the systems attracting set to be a NESS density, then its existence will entail a generative model. This is a result of the statistical generalisation of the complete class theorem. Thus in selecting to model a system under the FEP, we have presumed its dynamics to entail a generative model. This says nothing, however, about any empirically-ascertainable properties of living systems.

Both the literature forwarding the FEP and the literature critiquing it suffer from issues of translation: would-be interpreters of the framework face the burden of translation between linguistic and mathematical descriptions, between discrepancies in disciplinary standards and terms of art, and the long history of translation between various applications and interpretations which components of the mathematical framework itself have undergone. I urge that by unpacking these discrepancies in disciplinary conventions, scrutinizing the history of the formalism, and divorcing it not only from meanings ascribed to it in past disciplinary settings and applications, but from meanings applied to it in what I term "FEP models," operationalisations of the framework, we can come to a considered understanding of what the FEP is in its own right.

1.5 Models & The FEP

1.5.1 The FEP as Model Structure

I propose here that we reserve "the free energy principle" to denote only the maths of which the FEP is composed. Models utilizing this formalism to study natural systems bear also an interpretation, lending a means of interpreting the maths as *about* systems in nature. The existing literature on the FEP, however, nearly all seems to leverage the FEP to address some issue, relating it to a target system, producing what we might call "FEP models."

According to Weisberg's (Weisberg, 2007, 2013) account of models, a model's structure—whether mathematical, computational, or physical—does not inherently relate to a target system in an epistemically fruitful way, e.g., by representing features of that target system. It is only with the addition of a scientist's interpretation or construal that we derive a mapping between a model structure and the world, and a model is born. Once we have that model in hand, however, it can be tempting to say that we have knowledge of the natural world. The existence of models, their features, and the output of modeling work, however, do not constitute knowledge of nature over and above empirically-observed facts that we may have plugged into our models at the outset (if they are the type of models that incorporate measurements). To validate a model or its results and derive from it knowledge of natural laws, systems, or processes, we must match its predictions to experimental evidence.

We make a category mistake when we claim that a raw mathematical structure lends us predictions or places constraints on what can be observed in nature, and are guilty of reification: "there is no such thing as a solely mathematical account of a target system" (Nguyen & Frigg, 2017, p.1). Likewise when we take the existence or qualities of a model to constitute knowledge of the natural world we make a category error and reify the model.

Models have been proposed utilizing the formal structure of the FEP that target various domains, including cortical structure and neuronal organization (Friston et al., 2021), the brain as a whole (Friston, 2010), organisms acting in an environment (Bruineberg and Rietveld, 2014; Bruineberg et al., 2016), morphogenesis of multicellular organisms (Kuchling et al., 2019, 2020), and even social structures and behaviors (Ramstead et al., 2016). These models encompass representationalist FEP approaches (Kiefer and Hohwy, 2018, 2019), cognitivist FEP models(Kiefer and Hohwy, 2018, 2019), non-representational, Gibsonian FEP models (Bruineberg and Rietveld, 2014; Bruineberg et al., 2016), enactivist FEP models (Kirchhoff and Robertson, 2018; Ramstead et al., 2016), as well as both dualist (Hobson and Friston, 2014) and materialist (Friston et al., 2020b; Kiefer, 2020) FEP models. Each of these is referred to in the literature as "the free energy principle," with the ontologies, epistemologies, and predictions borne of each deemed consequences of "the free energy principle."

That one and the same "theory" can lend itself explanations of neuronal, cellular, and social organisation seems puzzling; that it can lend support for both neurocentrism and extended cognition even more so. I propose that we can resolve the source of this error signal by denying, in the first place, that the FEP is a theory—or even a model. Instead, we ought to use "the FEP" to denote the model structure: the raw formal framework, sans interpretation. In combination with the numerous construals that exist in the literature, this structure becomes "FEP models." The FEP itself, then, lacks all empirical or conceptual content. It is an empty formalism. The many models built thereon, however, can be seen to differ with respect to their content (thus different targets, conflicting interpretations of the same target). If we continue to take the FEP to refer to all of the above, then the conclusion we must reach (and that critics of the framework have reached) is that the FEP must either be vacuous or internally inconsistent. I say we bite the bullet on vacuousness, but restrict the term to be used only in reference to the formalism.

To illustrate briefly what I mean here: Ramstead et al. (2018) write that: "The FEP is a mathematical formulation that explains, from first principles, the characteristics of biological systems that are able to resist decay and persist over time," (p. 2) and that it "asserts that all biological systems maintain their integrity by actively reducing the disorder or dispersion" (p. 3). I wish to urge that claims to the effect that the FEP "explains" or "asserts" anything are misguided. Rather it is only FEP based models which can assert or explain and, indeed, relate at all to the world. Compare this to Kiefer (2020) who more rather more carefully describes his work as a "conjunction of the free energy principle...and the identity thesis" (p.1). Here I am not making a prescriptive case that we ought to always and only ever refer to "the FEP" as the maths in the absence of any interpretation and to the models constructed therefrom as "FEP models," though admittedly this might go a long ways in clearing up some of the misapprehensions in the literature. Rather it is my hope that this distinction will equip the casual reader with the conceptual tools necessary to parse the FEP literature, as notoriously dense and befuddling as it is.

The "FEP" as it has been addressed in the literature thus far is something of an impossible figure: A pure formalism, empty of all content. A precise theory of neuronal signalling and transient ensembles. A tautology. A transcendental argument. It is both unfalsifiable and yet makes precise predictions about the physical or physiological systems and dynamics capable of instantiating it. Both materialist and dualist. Both representational, cognitivist, and neurocentrist and, at the same time, ecological, enactive, and extended. In this section, I have argued that the only way around interpreting the FEP as over-committed to conflicting claims is to distinguish the formal framework from various models composed therefrom.

We ought, then, to think of the FEP as a mathematical structure alone, free of conceptual content, predictions, or representations of worldly systems. Models add to a structure an interpretation, or construal, which lends them a mapping function to worldly systems. Models composed from the formal structure of the FEP may or may not make predictions, or place constraints on the varieties of systems capable of realizing the dynamics they specify. This is likely to differ from model to model. Attempting to saddle the formal architecture of the FEP itself with falsifiable predictions, however, is simply a category error.

1.6 Conclusion

A recent and abundant literature concerns itself with the FEP, though its claims and status are contested. The FEP as it is addressed in this literature—both for and against—is a mathematical structure applied to the modeling of various phenomena across the social, cognitive, neuro-, and life sciences. Attempts to secure the precise nature of the FEP, its utility for scientific practice, just how it represents systems in its respective domains of application, how precisely it is bolstered or refuted by existing empirical evidence, what constraints it places on process theories and lower-order models, have thus far been foiled. I suspect that this is the case because the questions we have been asking of the framework make implicit category errors, attempting to saddle it with attributes that fail to apply to the framework in virtue of the sort of thing that it is. What are the FEP's theoretical commitments? What empirical support does it receive? There are as many answers to these questions as there are papers on the FEP, and continuing to ask them of the modeling framework as a whole, without regard to the distinct forms it assumes, will remain a fruitless undertaking.

In this chapter, I have argued the case that the free energy principle be considered not a theory, a law, a hypothesis, a paradigm, or a model, but a formal modeling structure. One immediate consequence of this conclusion is that questions of the epistemic status of FEP models, their empirical content, the predictions they do or do not make, and their precise relation to various corollary models and process theories will have to be assessed piecemeal, for each FEP model in its own right. This seems to pass the explanatory buck. In this respect, the state of the FEP mirrors the state of the modeling literature at large: there is very little that can be said evaluatively of models as an undifferentiated whole. There are, though, broad-brushstrokes appraisals to be made of the sorts of models that emerge from the FEP. Following this, there are more exacting claims to be made about specific instantiations of the framework. These, however, will have to be the subject of later work.

Another, more positive consequence, however, is that, having separated the formal essence of the FEP from various interpretations thereof, we are now free to build theoretical or empirical models with it without worrying about its theoretical commitments or empirical support, for it has none. The only barrier to utilizing the FEP is understanding the maths and understanding how to relate it to an open question or target system of interest. We did not enquire as to the theoretical commitments or empirical support of the evolutionary algorithm or Conway's game of life; we simply played with them. Our approach to the FEP ought to be the same.

That no silver bullet will vanquish—or vindicate—the spectre of the FEP once and for all may come as a disappointment to some. Have all our attempts to nail it down been in vain? I, for one, think not. It is my sincere hope that one positive result of this exercise will have been to disabuse a few scholars of outmoded conceptions of the scientific method. Scientists are not everywhere all the time dealing in literally true direct representations of natural systems. Arguably, much of what we are doing as scientists—especially now, especially in the younger disciplines—is far more heuristic and bottom-up than the theory-driven, hypothetico-deductive, falsificationist frames of yore would have us believe.

It is perhaps natural—at least not uncommon—to think that without hard and universal desiderata for what differentiates scientific pursuits from other intellectual projects, for what separates good and bad science, that we will slip into everything-goes relativism or pluralism. An effort is only scientific, and hence, worthwhile, if it makes concrete, falsifiable predictions. If we adopt this frame, most of today's scientific methods are not scientific at all. In fact, many of our historical bastions of the scientific method have failed to live up to these rigorous standards. It should not be considered a failing, however, that Galileo, Newton, and Darwin's work failed to be paradigmatic, failed to be theory-driven, failed to uncover mechanisms, or failed to conform to a hypothetico-deductive model. Model-based philosophy of science embraces the messiness and pluralism of scientific practice. It has, at times, done so at the cost of being overly-permissive. Pluralism as a thesis, however, does not absolve us of the responsibility of distinguishing good and bad science, working methodologies from those that have become enmired. That we cannot dismiss the FEP outright for failing to put forth a testable hypothesis—or accept it because it purports to explain a great many things—should not cause us to throw our hands in the air. It should be, rather, a call to arms, an impetus to get ever more exacting.

CHAPTER 2

SCIENTIFIC MACHINE LEARNING & THE THEORY-FREE IDEAL

2.1 Introduction

This chapter contends with the notion that the methods of machine learning (ML) are unique among the tools of science in enabling a form of theory-free inductive inference. I contest these assertions of the epistemic distinctness of ML, attributing the prevalence of these views to an untenable conception of scientific objectivity: what I term a *theory-free ideal*, in homage to its normative counterpart. By means of two case studies, I argue that this theory-free ideal has a deleterious effect on the epistemic standing of ML-involving science.

In the decade elapsed since the deep learning revolution, machine-learning (ML) techniques have established a firm foothold throughout the sciences. Successes of such methods have ranged from real-time particle-event sortation at the Large Hadron Collider (Duarte et al., 2018) to DeepMind's Nobel prize-winning accomplishments with its AlphaFold2 system for protein-structure prediction (Jumper et al., 2021). According to certain spokespeople from science and engineering communities, however, these achievements are trivial in comparison to the potential such methods hold for scientific advancement.

Various commentators have asserted with some frequency that ML will instigate profound—even "revolutionary"—changes to the nature of science and the knowledge it produces (Anderson, 2008; Boge, 2022; Hey et al., 2009; Mayer-Schönberger and Cukier, 2013; Society and Institute., 2019; Spinney, 2022; Srećković et al., 2022). The people behind such claims see ML methods as holding the potential to retire or else displace the role of theorizing in science (Anderson, 2008; Mayer-Schönberger and Cukier, 2013; Spinney, 2022; Srećković et al., 2022). Desai et al. (2022) refer to this conception of an ML-enabled scientific paradigm as "the epistemically revolutionary new frontier raised by data science: the so-called 'theory-free' paradigm in scientific methodology" (p.469). Some of these statements regarding the scientific usage of ML echo proclamations that were once made of domain-generic statistical analyses: Levins and Lewontin (1985) write of the motivation behind principle component analysis (PCA) and regression techniques that researcher's "assumption is that they are approaching the data in a theory-free manner and that data will 'speak to them' through the correlation analysis" (Levins and Lewontin, 1985, 156). We see this sentiment echoed today in the assertion that big data analytic tools promise to allow the raw data to "speak for themselves" (Anderson, 2008, 1).

This theory-freedom is intended as a negation of theory-mediation, theory-drivenness, theory-involvement, and theory-ladeness. It is also, we will see from an examination of the source literature, a denial that methods rest essentially on domain-knowledge or prior conceptualization of the target phenomena, or that they should be understood as representing features of target systems in any inference-licensing respect. "Theory" is hence to be understood in a broad and colloquial sense, as incorporating domain knowledge or conceptualization of target phenomena. Subscribers to the theory-free ideal seek to purge science of what they see as epistemically compromising arbitrariness and subjectivity. This subjective element is brought on board when human critical thinking or conceptualization of target phenomena play an essential role in shaping an empirical research program.

Taken at face value, the claim that ML could enable a form of theory-free inductive inference—or even inductive inference that differs qualitatively in its degree of theoretical support—does not withstand scrutiny. That inductive inference rests on pregiven conceptual infrastructure is, I take it, an effective premise of all discussions of the subject since Hume. Neither does inductive inference move along a sliding scale between theoretically and empirically driven: induction is, definitionally, a form of inference that requires both theoretical and empirical support. These are neither quantifiable nor scalar.

Ample scholarship in epistemology dating back centuries has characterized induction.

The present chapter is not a thesis in epistemology, nor is it intended as a revision to philosophical conceptions of induction, of which I take Hume's characterization to be adequate. My aim in this chapter is, instead, to analyze claims regarding the use of ML in science, its epistemic status, and its transformative potential. My engagement with these questions seeks to improve upon prior philosophical treatment in distinguishing between normative and descriptive agendas. Claims of the epistemic distinctness of ML, I contend, latch onto real novelty in some instances of ML deployed toward scientific ends: potential for misuse and lack of methodological standards. Instead of identifying this as the epistemic problem it represents, however, claims of epistemic distinctness and theory-freedom function to reify the (potential) misuse of ML-based tools into an account of how these tools normally function, how they necessarily function, or even how they normatively *ought* to function.

In the course of this chapter, I will characterize the misdirected conception of scientific objectivity implied by these claims. I refer to the meta-narrative that endorses this notion of objectivity as a *theory-free ideal*, paralleling the *value-free ideal*, which is its normative counterpart. This meta-narrative, I will show, is detrimental to the epistemic soundness of science conducted with ML.

My argument proceeds as follows. Claims of the epistemic novelty of ML-involving methods concerning their reliance on theory have been alleged by both scientists and philosophers. I furnish exemplars of these in 2.2. Successful inductive inference, however, requires background assumptions; this is one of the defining characteristics of inductive inference, according to known formal and epistemological accounts, as I show in section 2.4. This presents us with a dilemma: either revise our understanding of induction, or conclude that ML only enables inductive inference in virtue of theoretical input. I urge that the effects of such beliefs on scientific practice with ML should be the most salient consideration; we should accept the second horn for pragmatic reasons, which I endeavor to motivate via case study. In Section 2.5.1, I show that when ML is incorporated into scientific pipelines to epistemic success, it is in virtue of working explicitly with theoretical resources. These come in at the junctures of problem formulation, data collection and curation, model design, model training, and model evaluation. When, in contrast, investigators ignore or suppress theoretical assumptions in pursuit of an ideal of theory-freedom, methodological pathologies emerge: statistical artifacts are mistaken for structure, arbitrary modeling choices or starting assumptions are mistaken for empirical discoveries, and downstream inference is rendered unreliable. A case study explored in Section 2.5.2 illustrates this contrast class. I attribute the prevalence of the theory-free conception of ML to a meta-narrative concerning ideals of scientific objectivity: the *theory-free ideal*. I conclude with recommendations for scientists in working explicitly with theoretical resources, and recommendations for philosophers in their engagement with ML and the claims surrounding it.

2.2 Distinctness

2.2.1 The beliefs of working scientists

A monograph titled "The AI revolution in scientific research," released jointly by The Royal Society and the Alan Turing institute, offers scientists' own assessments of anticipated changes to scientific practice spurred by the involvement of ML (Society and Institute., 2019). Summarizing the opinions of the assembled scientists, the authors write that "AI" is set to have "a disruptive influence on the conduct of science"(Society and Institute., 2019, p.10). Such pronouncements appear to be underpinned by a conception of the workings of ML in science as a theory-free enterprise, given the authors' description of the normal function of ML and data scientific methods. The standard way to apply ML in science, they write, is "to start from a large data set, and then apply machine learning methods to try to discover patterns that are hidden in the data—without taking into account anything about where the data came from, or current knowledge of the system" (Society and Institute., 2019, p.9). The authors explicitly contrast this use case with the potential for more theory-driven research techniques, à la PINNs (physics-informed neural networks). However, it is clear from the exposition that a theory-agnostic conception of the typical function of ML models informs the authors' predictions of disruption.

Chubb, Cowling, and Reed (2022) conducted a survey of identified leaders across various scientific fields concerning the adoption of AI/ML based methods within their research practices. A consistent theme amongst the researchers surveyed was the sentiment that "AI could prompt 'unforeseen' outcomes, potentially leading to a reframing of disciplines, modes and methods of knowledge production" (Chubb et al., 2022, 1442), and that "AI could be used in the near future to bypass traditional means of knowledge production" (Chubb et al., 2022, 1445). One interviewee explained the difference between "traditional" and AI-based methods as follows: "[n]ormally the scientific progress goes like this, so you have a hypothesis and then you collect data and try to verify or falsify the hypothesis, and now you have the data and the data, so to say, dictates you what hypothesis you can find. So, this is how methodologies, scientific methods are changing" (Chubb et al., 2022, 1446).

These overviews of scientists' perceptions of the place of AI in science, and its potentially transformative role, paint a relatively coherent picture. Machine learning, or "AI," enables scientists to carry out their work in a far more data-driven, and far less theory-driven capacity. Certainly, some research paradigms (or stages within a research pipeline) are more exploratory than others. A distinction between exploratory (broadly, data-driven) and explanatory (broadly, theory-driven or theory-involving) research strategies is popularly held by working scientists. The picture these assembled voices paint, however, seems to point to a lessened overall need for theoretical input within scientific discovery.

Articulations of the distinctness of ML emanating from science journalists à la Anderson (2008), (Hey et al., 2009), Mayer-Schönberger and Cukier (2013), and (Spinney, 2022) no doubt represent far more sensationalist visions for the role of ML in science than most working scientists would assent to. The average scientist would likely deny that AI/ML will soon altogether obviate the need for theory, preconception, or domain-expertise within scientific knowledge-production. Nevertheless, the overarching perception that the methods of science can or should be rendered free from theory exerts a force on the research practices of working scientists. Funding for grants and for institutes and centers, as well as industry sponsorship for conferences, awards, and the like often hinges on scientists conveying the novelty and disruptive potential of their methods which, increasingly, is tied to an ideal of theory-freedom.

2.2.2 A philosophical defense

Philosophers have been quick to respond to assertions that the rising tide of ML-adoption will enable a "post-theory science" (Spinney, 2022, 1). Some philosophers have critiqued this vision of ML-infused science, some endorsed it, while others have simply acknowledged its ubiquity (Alvarado and Humphreys, 2017; Beisbart and Räz, 2022; Boge et al., 2022; Boon, 2020; Creel, 2020; Desai et al., 2022; Duede, 2023a; Hansen and Quinon, 2023; Kawamleh, 2021; Kitchin, 2014; Leonelli and Zalta, 2020; Pietsch, 2021, 2022; Pigliucci, 2009; Rowbottom et al., 2024, 2023; Sullivan, 2022; Srećković et al., 2022).

Alvarado and Humphreys (2017) take stock of observations on ML and big data from scholars hailing from a range of disciplinary backgrounds. These scholars describe the widespread adoption of ML and "big data" analytic methods resulting in "a common epistemological effect" (Alvarado and Humphreys, 2017, 739). The primary manifestation of this "epistemological shift" being that "[t]heory...at the level of how knowledge is produced and structured...[has] been replaced by information stored in databases too large to read and processed by algorithms too complex to understand"(Alvarado and Humphreys, 2017, 739). If ML or big data analytic methods are indeed "interpretation-free," Alvarado and Humphreys write, this will entail "a permanent change in the way that science is pursued"(Alvarado and

Humphreys, 2017, 744). In a treatment of the representational status of ML in science and its relation to the scientific realism debate, Rowbottom, Curtis-Trudel, and Peden (2023) begin from the premise that scientific ML "contrasts with traditional scientific modeling, where explicit theories and models are used" (Rowbottom et al., 2024, 172).

In a 2021 paper, Duede writes that philosophers and scientists alike have widely made claims of the epistemic distinctness of ML and its disruptive potential. Duede observes that "to scientists and science funding agencies alike, artificial intelligence both promises and has already begun to revolutionize...science" and that "nearly every empirical discipline has already undergone some form of transformation as a result of developments in and implementation of deep learning and artificial intelligence"(Duede, 2023a, 1089). But, as Duede notes, philosophers and scientists, while agreeing on the revolutionary potential (or actuality) of AI/ML in science, have made separate meaning of it. Duede sets out to address these discrepancies, attributing what he perceives as philosophical pessimism concerning the role of ML in science, in large part, to "a failure on the part of philosophers to attend to the full range of ways that deep learning is actually used in science"(Duede, 2023a, 1090). In his critique of philosophical reactions to claims of the novelty of ML, however, Duede leaves these theses unchallenged. I will argue that the failure Duede documents on the part of philosophers to account for how ML might actually be implemented in scientific practice is ultimately responsible for philosophical endorsement of claims of ML's distinctness.

Srećković, Berber, and Filipović (2022) differentiate machine learning techniques from standard practices in statistical modeling, arguing that statisticians employ theoretical assumptions, while machine learning practitioners do not. "ML models," the authors write, "are constructed based on data instead of theoretical assumptions about the target system.(Srećković et al., 2022, 166).

Srećković, Berber, and Filipović (2022) evaluate what they hold to be the key differences between traditional modeling approaches and machine learning methods in terms of the explanatory capacity of both and their ability to elucidate causal relationships. Srećković et al. diagnose the methods of machine learning as uninterpretable, and not resting on theoretical considerations. This, according to the authors, prevents the practice from getting at underlying causes and furnishing explanations of natural phenomena. The ability of ML techniques to provide prediction in the absence of explanation is projected by the authors to alter the landscape of how we conduct science.

"In contrast to explanatory-focused statistical models," Srećković et al. argue, "ML models reach predictions without the theoretical backup that supplements the correlations found in the data with a potential causal interpretation" (Srećković et al., 2022, 160). Machine learning, they argue, is "theory-agnostic" in that "there are no a priori assumptions concerning the mechanism of the target phenomenon" (Srećković et al., 2022, 165). While the authors acknowledge a sort of disappearing line between ML and traditional statistical techniques, their emphasis is on drawing out broad characterizations of the two disciplines and what separates them. Whereas for "traditional statistics, standard models rely on the representation of underlying causal mechanisms, and they are used for retrospective testing of an already existing set of causal hypotheses...ML models are constructed based on data instead of theoretical assumptions about the target system. The purpose of such models is primarily forward-looking, i.e. to predict new observations" (Srećković et al., 2022, 166). Here, the contrast the authors draw between broadly "data-driven" and "theoretically-motivated" methods is telling. This distinction is not one the authors have introduced: such a divide between theory-driven or hypothesis-driven research and data-driven research is held widely among engineers and scientists. Srećković et al. merely provision a philosophical exposition and justification thereof.

In a similar vein, Boge (2022) speculates that a revolution in either scientific practice or its epistemic footing may be in store owing to the adoption of machine learning—specifically deep learning—methods. Boge's argument rests on the idea that deep learning is both instrumental in an idiosyncratic sense among modeling approaches in the sciences, and that it exhibits a novel kind of epistemic opacity to its deployers. These identifying facets of deep learning pose an impediment to understanding and explanation (in the scientific sense), especially when deployed in exploratory settings where the successful results of scientific enquiry will require novel concept-formation. Owing to their divergence from standard mathematical modeling practices in the sciences, Boge claims, ML modeling techniques “have the potential to profoundly ‘change the face of science’” (Boge et al., 2022, p.71).

Boge urges that the distinction between the procedure of classical mathematical modeling or computer simulation in science and the application of machine learning methods is that the former procedure begins with a conceptualization of the target phenomenon under investigation, while this step is absent in the use of ML. “The difference,” Boge writes, “between CS [computer simulation] and DL [deep learning] may be summarized as follows: The former begins with a conceptualization of the target, and from that predicts ‘hypothetical data’. The latter begins with a conceptualization of data” (Boge et al., 2022, p.59).

Especially in exploratory modeling contexts, the lack of background theory or conceptualization of the target phenomena is taken, by Boge, as a potentially serious impediment to understanding. While Boge grants that DL models might represent, he holds that they fail to be explanatory for lack of theoretical context and conceptual content, writing that a “DL model...is conceptually too poor to provide an understanding of underlying mechanisms” (Boge, 2022, 57). Boge takes after de Regt in his stance on the relation between representational status and explanatory status: “for representational models to explain, they must also be constructed under the principles of an intelligible theory, where a theory is intelligible if it has certain qualities that ‘provide conceptual tools for achieving understanding’ (de Regt, 2017, p. 118)”(Boge, 2022, 54). Boge predicts profound changes to the practice and epistemic products of science because ML-based tools will fail to provide understanding or explanations due to their lack of theoretical or conceptual motivation and content.

Boge and Srećković et al. both appear to sign onto the thesis that ML methods are theory-free or devoid of some essential variety of conceptual content which enables them to serve classical explanatory or inferential roles in science. The methods of ML are, hence, understood as distinct from canonical modeling methods in science and traditional statistics. Boge and Srećković et al. further contend that the widespread adoption of ML methods will catalyze disruptive change in science, while Boon argues that the theory-freeness of ML methods rules them out as viable tools for science. These scholars take the perceived differences between "normal science" or even "real science" and machine learning to amount to the degree to which they are theory-laden, theory-driven, or conceptually rich. As I will demonstrate in the subsequent sections, no use of ML in science is "theory-free," and those that aspire to this ideal tend to result in poor scientific practice.

Boon (2020) signs onto the distinctness thesis, maintaining that machine learning methods are in a category apart from classical statistical or scientific methods owing to their theory-agnosticism. "Machine learning," Boon writes, "is different from computer simulations, which utilize scientific knowledge to build mathematical models...[t]he machine-learning process does not draw on scientific models that are constructed by means of theories, laws, mechanisms and so forth. No theory or mechanism or law needs to be fed to the machine-learning process" (Boon, 2020, 47). Interestingly, she diverges from the norm in taking the distinctness thesis as a reason to *reject* the notion that ML will have a transformative influence on the conduct of science. Boon denies that machine learning methods will obviate the need for human conceptual apparatus in the generation of scientific knowledge, arguing science to be an essentially theory-involving activity. "[S]cience," Boon writes, "...cannot be replaced by machine learning technologies whatsoever since incomprehensive, opaque data-models do not tell us anything meaningful about the world. Therefore, 'real science' and machine learning technologies operate in very different domains and must not be regarded as competing" (Boon, 2020, 58).

Boon, Boge, and Srećković et al. each sign onto the idea that ML methods are in some sense theory-free or devoid of conceptual content, and hence distinct from canonical modeling methods in science and traditional statistics. Boge and Srećković et al. further contend that the widespread adoption of ML methods will catalyse disruptive changes to science, while Boon argues that it is precisely the theory-freeness of ML methods which rules them out as capable of unseating existing modes of knowledge-production in the sciences. These scholars take the perceived differences between "normal science" (or "real science," as Boon puts it) and machine learning to amount to the degree to which they are theory-laden, theory-driven, or conceptually rich. As I will demonstrate in the subsequent sections, no use of ML in science is "theory-free." Scientific applications of ML that aspire to this ideal of theory-freedom tend to result in poor scientific practice.

2.3 Conceptions of scientific objectivity

The concept of objectivity is central to modern science, both as abstract ideal and as a set of human practices. What variety of objectivity scientists ought to strive for has been contested territory for centuries. One thread of this debate concerns the extent to which scientific practices and the knowledge that they produce are ineliminably structured by human values. Another concerns the extent to which such practices and outputs are necessarily structured by theory, in the sense of conceptual content, or prior commitment to the nature of the subject-matter.

Philosophical conceptions of objectivity are rooted in accounts of the nature and possibility of empirical knowledge. They are highly abstracted from on-the-ground empirical practices and have little direct influence on them. But scientists in modernity have operated with their own, albeit often implicit, conceptions of scientific objectivity. These have permeated public conceptions of science which, in turn, feed back into scientists' self-conceptions of their work and its epistemic foundations. Thus philosophical conceptions of scientific objectivity and meta-narratives of scientific objectivity come apart.

A recent literature on values in science has offered an extensive treatment of the philosophical conception of objectivity as freedom from normative influence and its corollary meta-narrative: the value-free ideal. Few historical interlocutors have put forward explicit defense of objectivity as total value-agnosticism. Instead, it has been argued that science strives to minimize the impact of human values or to constrain their influence to appropriate venues and junctures. A mostly implicit ideal of total value-freedom, however, is widespread and influential. Its influence extends to scientific practice, to public reception of science, science education, and to the interplay of science and public policy (Douglas, 2009). Philosophers of science have argued that the end goal of total freedom from normative influence is unachievable on both practical and in-principle, epistemic grounds (Douglas, 2009; Elliott and McKaughan, 2014; Longino, 1990). Denying the necessary influence of values on science, it is argued, merely cements them, lends them an air of objectivity, and renders them unavailable to critical scrutiny.

So then, there is a broadly philosophical doctrine which conceives of scientific objectivity as a minimization of the undue influence of normative values on science. This doctrine comes apart from a scientific meta-narrative which pushes for the total elimination of values in science. This meta-narrative, according to feminist philosophers of science, is neither achievable nor desirable. I take it to be important to make the distinction here between the well-reasoned doctrine which has had actual historical proponents[1] and the harmful meta-narrative. Strains of the values in science literature have had a habit of slurring these together.

For our purposes, we can remain agnostic as to the truth of either the philosophical doctrine on values in science or its parallel metanarrative. Here we are concerned, instead, with a conception of scientific objectivity centered on the appropriate purview of theory. Much

1. For instance, Max Weber.

like the value-centered conception, there is both a reasonable doctrine and an extremefied meta-narrative. The doctrine calls for scientists to constrain the influence of theory to appropriate venues. After all, we do not want our research efforts to be artificially constrained by what we think we know about the phenomena, what we have conjectured about the phenomena, or simply the limitations of our theoretical apparatus. Essentially, this doctrine calls for our research practices to be *non question-begging*.

What I am calling the *theory-free ideal* is the corresponding meta-narrative. According to the theory-free ideal, science should strive to minimize, or even eliminate theoretical input from empirical research. With the advent of ML-assisted science, belief in the narrative of theory-freedom has become commonplace. Leonelli (2020) observes that one of the dominant responses to the rise of ML and big data analytic methods in science is to see it as a championing of what I have here dubbed the theory-free ideal: "[one] way to interpret the rise of big data is as a vindication of inductivism in the face of the barrage of philosophical criticism leveled against theory-free reasoning over the centuries" (Leonelli and Zalta, 2020, Sec.6, Par.4). The meta-narrative that tells us that science can be rendered theory-free is not innocuous: it effectively serves to conceal loci of theoretical input and reifies the implicit beliefs of uncritical scientists.

No doubt, the deep incorporation of ML methods into empirical research pipelines brings about changes to where domain knowledge and theoretical considerations come to bear on the scientific process and its outputs. The case studies reviewed in Section 2.5 are revelatory of some of these differences. Fundamental changes to the nature and loci of theory-impingement, however, have occurred continuously throughout the history of science. The development of computer simulation, sampling methods, or the formal apparatus for statistical analyses essentially shifted where theoretical considerations came into play in the inferential process. So, too, for that matter, did the Newtonian style of mathematical thought-experimentation and his method of fluxions. Novel conceptual tools entail novelty to the nature of conceptual influence on the brute work of empirical inference. None can obviate the need for conceptual infrastructure, nor can they open up novel pathways to knowledge of the world.

2.4 The necessity of theory

2.4.1 In induction

Inductive inference is a form of inference grounded in empirical observation. Induction is contrasted against deduction, and lacks the guarantees of deductive inference. The 18th century philosopher David Hume's several treatises on the subject lend us our modern philosophical conception of induction through a skeptical appraisal—what has come to be known as the problem of induction. As Hume motivated the puzzle, there is no frequency of occurrences of an identical phenomenon that would justify inference to an inductive generalization with certitude. No matter on how many occasions we have seen the sun rise, we are not licensed to the certain knowledge that it will rise again tomorrow. Inductive generalization, then, is a means of arriving at knowledge requiring infrastructure that goes beyond a mere spate of observations. Inductive generalization requires background conceptual infrastructure, or theory, to get off the ground. Indeed, the very ability to categorize several experiences as "instances of the same phenomenon" requires a concept or "theory" of the phenomenon. Such classifications are, of course, essential to inductive generalization. That conceptual or theoretical resources must be brought to bear on an inference procedure to license inductive generalization is essential to our modern philosophical understanding of induction.

2.4.2 In science

The work of science produces empirical knowledge by means of inductive inference.[2] Induction, in turn, rests on theoretical resources. Sellars (1956)'s characterization of the "myth of the given" offers an account of the necessity of conceptual frameworks in the act of observation and inductive generalization, without which science could not generate empirical knowledge. Norton (2003)'s "material theory of induction" describes how successful inductive inference is never licensed by universal, domain-generic formal rules, but always proceeds by the application of local rules warranted by hard-won empirical—in Norton's words, "material"—facts tied to a specific scientific research context (Norton, 2003).

Turning to scientific practice, even simplistic experimental designs reveal the nature and extent to which scientific observation and inference are theory-inflected. The very act of investigation involves commitment to the existence and in-principle measureability of some phenomenon. If we are making measurements and performing quantitative analyses thereon, we are further committed to the phenomenon being amenable to quantitative representation and analysis. How we choose to measure and analyze records of a phenomenon generally includes a commitment to its quantitative ontology, e.g., is it categorical, ordinal, or cardinal? Measurement cannot be total, and therefore there is always a commitment as to what to look at experimentally and what to exclude. The very design of our instruments of measure and their calibration includes various commitments to the nature of the worldly phenomena under investigation. There is always, for instance, a commitment to the appropriate level of abstraction at which to study the phenomenon in question, which manifests in settings on instruments of measure, such as degree of magnification or periodicity of sampling. In fundamental physics, when we cool our instruments to reduce the contamination of our

2. While certain 20th century philosophers of science, including Hempel and Popper, made cases for the role of deductive reasoning in scientific inference, these projects are generally considered to have failed by their own lights. E.g., Popper's admission that some hypotheses could receive greater or lesser evidential corroboration pushes him to an inductive view of scientific inference.

measurements by thermal noise, it is our prior theoretical grasp on the target phenomena, the physical systems under study, that motivates us to do so.

Crucially, "data" does not refer to physical phenomena.[3] "Data" refers to abstract, formalized representation of the results of direct observation or measurement. Data must be capable of serving an evidential role in licensing inferences about natural phenomena. Given that data is a form of mathematical representation, it does not intrinsically hold semantic meaning or refer to empirical phenomenon. The meaning that data holds for scientific inference exists in virtue of human interpretation and empirical grounding. For the use of any mathematical analysis—including the modes of analysis enabled by ML—to ground any scientific inference, it must be given conceptual content. This is already an essential form of theory-ladenness. The parameters of any machine learning model and its outputs are a step removed from input data, but are likewise mathematical representations. The data-derived parameter weights of a neural network, for instance, capture salient statistical patterns in the training data which are then leveraged to regress or classify the data on which they are tested or deployed. They represent abstract features of the training data. The representational status of neural network models is derivative of the representational status of the data on which they are parameterized.

A number of philosophers have provided strong rationales for rejecting the possibility of theory-free science. Leonelli (2012, 2018, 2020) stresses the essential theory-ladenness of data, decrying the popular conception of data as "raw" and "objective." Leonelli (2018) investigates "the different extents to which theory—understood broadly as a set of theoretical commitments and goals—impinges on inferential processes from data" (Leonelli, 2019b, 22). In several book-length treatments of the use and interpretation of data in scientific practice (e.g., (Leonelli, 2018, 2019a; Leonelli and Tempini, 2020; Leonelli and Beaulieu, 2021)), Leonelli concludes that there is no place in scientific practice in which we have data that is

3. See Sections 3.3.2 and 3.3.3

not already, to some degree, shaped by our existing conceptual or theoretical grasp on the phenomenon, commitments to epistemic goals and questions to be answered, idealizations, and auxiliary assumptions.

This view is a rejection of "[t]he naïve fantasy that data have an immediate relation to phenomena of the world, that they are 'objective' in some strong, ontological sense of that term, that they are the facts of the world directly speaking to us" (Longino, 2020, 391). Bogen (2016) argues that it is the very fact that data is not raw, that it is, in a sense, "impure" that makes it able to serve the meaningful epistemic role it does. Boyd (2018); Boyd and Bogen (2009) argues further that it is not in spite of, but owing to the theory-ladenness of data that empirical science garners us its epistemic results.

2.4.3 In machine learning

Inductive inference is the procedure of gaining knowledge by extrapolating from a limited number of observations to a more general class. The fundamental task of ML is the extraction of statistical patterns from a training dataset and the extrapolation of this pattern to prediction or classification tasks on unseen instances. ML is therefore, straightforwardly, a class of formal methods for inductive inference (Bergadano, 1993; Harman et al., 2007; Sterkenburg and Grünwald, 2021). If inductive generalization writ large cannot be accomplished without theoretical input, then no specific formal inference scheme can accomplish inductive generalization without theoretical input. Any ML-enabled inference, therefore, requires theoretical input—whether explicit or implicit. One of the primary places this theoretical input comes into play is in *problem formulation*, which includes the articulation of an inference task, the conceptualization of input data and learning objectives, and the election of success criteria.

We have discussed that inductive inference is data-driven, that scientific inference is data-driven, and that data, in these contexts, must be shaped by human concepts or theoretical resources in order to scaffold these inferences. The data that served to support inference in

ML is no different. Several accounts detail the role of theory in the variety of data on which ML is trained and deployed—often referred to as "big data."

Kitchin (2014) echoes that features of data collection and processing render data essentially theory-laden, in light of culturally-shared and ubiquitous background theoretical understanding of phenomena. Further, as Kitchin argues, data deprived of all semantic meaning would be uninformative, that is, unable to serve their essential epistemic role of scaffolding inference. In a similar spirit, Frické (2015) argues that theory must guide the selection of data to scaffold algorithm-assisted inference. Hansen and Quinon (2023) argue that ML-assisted science can never be made theory-free, as theoretical considerations necessarily enter in at the junctures of problem-formulation, data collection and curation, data pre-processing, and model-selection and validation. Desai et al. (2022) note that the theory-ladenness of observation makes it impossible to make observations or take measurements without the guidance of background theory. Desai et al. echo common sentiments among philosophers about the prospects of a wholly predictive science: such a view of the process of arriving at empirical knowledge is a naïve one, and ignores that one of the primary aims of science is explanation or understanding of the world.

Interestingly, it is not only philosophical considerations that lead us to the conclusion that ML-guided inferences cannot be rendered free from theory. Formal results from statistics and ML independently reveal the reliance of such methods on inductive biases or a priori constraints on the hypothesis spaces through which they search. Statistical learning theory (SLT) is the branch of theoretical computer science that looks to supply a theoretical basis (and formal guarantees) for inference with ML. The relation between ML/SLT as formal inductive method and the philosophical study of induction has not gone unnoticed: a number of scholarly works have treated the intersection of these subjects (Bergadano, 1993; Harman et al., 2007; Sterkenburg and Grünwald, 2021). These texts have drawn out the philosophical relevance of learning theoretic results for both the study of induction and the practice of ML.

The most salient results for our purposes are the no free lunch (NFL) theorems (Wolpert, 1996).

The no free lunch theorems are a set of results in statistical learning theory demonstrating the impossibility of a universally valid and purely data-driven inference rule. Though the philosophical implications of the theorems are vexed (see (Sterkenburg and Grünwald, 2021)), with some artistic license, we may think of the no free lunch theorems of supervised learning as a formalization of the idea that inductive inference would be impossible if the reality we inhabited (and, hence, the data we learn from) did not exhibit some (learnable) regularity (Wolpert, 1996).[4] In learning from data we must, therefore, make assumptions about the friendliness of that data to our epistemic intentions; empirical data alone is not enough to get induction off the ground (Sterkenburg and Grünwald, 2021). Learning from data is possible only with the incorporation of theoretical assumptions, in the form of prior selection of a model class, hypothesis class, or priors.

2.5 The role of theory in scientific ML

I have argued that the notion that widespread adoption of the methods of ML in science will obviate the need for theorizing is 1. widespread, 2. symptomatic of a theory-free ideal in science, and 3. untenable. In the final section of this chapter, I will attempt to illustrate its perniciousness by means of two case studies, which concern instances of actual application of modern ML methods in scientific practice. The first case study concerns a use case for ML in science that is deeply theory-laden and self-aware in its theory-ladenness. This use of ML in science has marked a scientific breakthrough, and been a resounding epistemic success. The second study concerns a use case for ML in science that is marketed as bypassing the need

4. More technically, the NFL theorems are an impossibility result for the existence of any learning algorithm that outperforms others on generalization to all learning environments, given the assumption of a uniform probability distribution over all possible learning environments. This uniformity assumption is, in a sense, an inversion of Hume's "uniformity of nature," which is the stipulation of structuredness to reality/experience/data.

for theory. This application of ML has been decried as statistical malpractice, its results at best uninformative, at worst, dangerously misleading. With these cases I aim to show the unavoidability of theoretical work in scientific applications of ML, and the deleterious effects of the ideal of theory-freedom on scientific practice.

2.5.1 The unreasonable effectiveness of AlphaFold

Perhaps the most impressive result that ML methods have achieved for science to date is AlphaFold 2.0. To appreciate the unprecedentedness of the AlphaFold results, we must first appreciate the scientific problem it is confronted with. The problem of protein folding is notoriously difficult. There is very little that we can say from the genotypic specification of a particular protein about how it will fold. Mapping from sequences of adenines, cytosine, guanines, and thymines to a menagerie of amino acids is relatively straightforward, as biological problems go; so is predicting the polypeptide chains these amino acid sequences will form. What mess of three-dimensional spaghetti those amino acid chains will assume once synthesized, however, is another matter entirely. This is an essential problem for the biomedical sciences. The three-dimensional anatomy of protein structure determines its function and is thus a crucial object of scientific inference.

To truly comprehend the difficulty of the protein folding problem—and how the methods of machine learning were able to get around it—we first have to recognize that protein structure is understood at four levels. DNA is a string composed of four alternating base pairs. It encodes information in sequence. When proteins are assembled, that DNA is read, codon by codon, and a polypeptide chain is built up from twenty amino acids on the basis of these instructions. These amino acid sequences are dubbed the "primary structure" of a protein. All amino acids are composed of the same base molecular structure, which will bond together to form the backbone of the polypeptide chain, containing an α-carbon, an amino group, a carboxyl group, and a hydrogen. From this molecular backbone extends the

R-group or side chain, the determinant of the amino acid's "flavor." The secondary structure of a protein refers to the morphology that polypeptide chains take on on their own, owing to bonding patterns in the backbone. The morphology of these peptide chains results from local interactions between adjacent and semi-adjacent molecules in the backbone of the peptide chain. Owing to the periodicity of the placement of amino acids with certain valences (and other molecular-bond determining features) in the chain, they will typically either form what are known as α helices or β sheets. Up until this point things have remained relatively straightforward, as biological problems go: we have a basic, repeated molecular structure and its self-interaction in the form of hydrogen bonding.

The tertiary structure of a protein is determined by the R-groups of the amino acids. Recall that these come in twenty flavors. Recall that virtually all forms of non-covalent bonding are available to these molecules now. Recall that amino acids can exhibit hydrophobic and hydrophilic proclivities. If a protein is composed of more than one polypeptide chain, it will have a quaternary structure as well. At the tertiary and quaternary levels of protein structure, we have advanced from assembling text from bit strings to attempting to predict all of the ways in which many distinct kinds of spaghetti thrown together in a pot can cohabitate, given six dimensions along which spaghetti substructures may or may not like to interact. Also, the spaghetti exhibits quantum behaviors.

At first blush, this seems like an unsolvable problem. The initial trick—the trick that gets existing bioinformatic solutions off the ground—lies in noting that when we have a variant in one amino-acid we can see what *non-local* variants tend to co-vary along with it. This begins to tell us something about what might be touching what in the tertiary and quaternary protein structures. Still a difficult problem, but more manageable.

The AlphaFold team began by creating their own sequence database—now the largest existing database of its kind—by large-scale clustering of existing sequence repositories. DeepMind's AlphaFold 2.0 takes an amino acid sequence as input in its pre-processing stage and derives a multiple sequence alignment (MSA). An MSA encodes evolutionary information, usually highlighting relationships of homology. In addition to the primary amino acid sequence and MSA, AlphaFold is also supplied as input database-derived *templates*—three-dimensional atomic maps—for a small number of sufficiently similar homologous protein structures. Distance and orientation features and sequence features are derived from preexisting template representations to form what the AlphaFold team dubs a *pair representation*, encoding relationships between pairs of amino acid residues.

AlphaFold treats the prediction of 3-dimensional protein structure from these pair representations and MSAs as a graphical problem, rendering the representations in the primary trunk of the model architecture into gradated bitmaps. The problem formulation for the DeepMind team was to "view the prediction of protein structures as a graph inference problem in 3D space in which the edges of the graph are defined by residues in proximity" (Jumper et al., 2021, 585). The core structure of AlphaFold 2.0 is a transformer: a form of DNN which exploits parallelization and attention mechanisms to incorporate contextual factors in the data at multiple levels of abstraction simultaneously—and, importantly, enabling information from these levels to 'talk to' each other. AlphaFold passes both the MSA and the pair representation (separately) back and forth through the trunk of the model for a set number of iterations (48 blocks) per recycle, progressively refining the representations, and allowing the two distinct representations (MSA and pair representation) to influence one another as each is refined. The output of this refinement procedure is then, in the final stage, fed to a structure module that uses Invariant Point Attention to output atom coordinates. The predicted coordinate map is then passed, with MSA and pair-representations, back through the trunk. This is repeated for three iterations until a final predicted 3D protein structure is achieved.

Let us draw out what is salient about this scientific procedure for our analysis, the aim being to examine the role played by theoretical considerations. The AlphaFold team explicitly incorporate theoretical resources at the stage of data provenance and engineering, the stage of architecture design, hyperparameter selection and model training, and at the stage of model evaluation and interpretation.

In the first place, theory integration comes in at the level of the data in terms of what the data ultimately represents and how it is imbued with that representational content. The data on which AlphaFold is trained is richly structured by existing empirical knowledge of the target domain (the molecular structure of proteins and their evolutionary trajectories) and our theoretical understanding thereof. AlphaFold sits atop a wealth of domain knowledge about the form and function of proteins. Theory also comes into play in how the data is handled for the specific task in question and how it is made to serve as evidence in this task. AlphaFold is, at its core, an instance of (semi-)supervised learning.[5] The exercise is premised on the idea that the rules of association between amino acid sequences and three dimensional protein structure lie latent in cross-taxa protein structure data. It is further premised on the supposition that the systematic breakdown in protein structure and function resultant from certain amino acid substitutions can be leveraged to learn the complex bonding affinities governing 3-dimensional protein structure. Part of what is noteworthy in this case study is the insight to take the publicly available data and turn it into novel representational forms in multiple places: combining MSAs and templates to create pair representations, and projecting those into effective heatmaps of sequence-structure associations so that the inference task could be treated like a graphical problem.

The architecting of the various model components utilized in AlphaFold 2.0 was similarly bound to theoretical considerations. AlphaFold is not a domain-generic model; the model architecture is hand-tailored to the specific task of learning to predict three dimensional protein structure from MSAs and pair representations—a novel representational form for

5. Supervised learning methods train a model to approximate a human categorization or decision, known, in ML, as a label. Unsupervised learning methods, by contrast, work to discover patterns from unlabelled data. Semi-supervised learning incorporates both labelled and unlabelled data.

the task. AlphaFold 2.0 employs a transformer network that is designed to iteratively refine progressively more accurate guesses at the true protein structure. The transformer trunk utilized in AlphaFold was created to combine and refine representations of the specific form it is fed in a novel training and deployment procedure. Perhaps the most strikingly theory-laden aspect of AlphaFold 2.0 is the engineering of specially tailored loss functions. In training a DNN, a loss function governs how the distance metric is calculated between present output and desired output of the model. In a typical neural network training regime, the error term is then backpropagated through the network, layer by layer, updating the weighting of the model's parameters so as to minimize the gradient of the loss. In specifying the loss function, machine learners are able to express precisely what it is that they are interested in learning for a particular task. In AlphaFold 2.0, the loss function is heavily tailored to the problem of predicting folded protein structure from amino acid sequences. The researchers employed "a loss term that places substantial weight on the orientational correctness of the residues" (Jumper et al., 2021, 585). Loss terms specific to the learning of various structural features of protein folding along a number of dimensions were employed at all stages of training and fine-tuning: "satisfaction of the peptide bond geometry is encouraged during fine-tuning by a violation loss term" (Jumper et al., 2021, 586-587).

Finally, model-evaluation, that is, judging the success of the trained model and interpreting its results requires integrating the resulting predictions of AlphaFold into existing biological knowledge. We can only judge the success of such a model when it is understood against the backdrop of our prevailing scientific accounts. AlphaFold 2.0's success was only legible in the CASP14 (Critical Assessment of Structure Prediction) experiment in achieving at or near the performance of theory-guided and experimentally-obtained protein structures. We can likewise only put the results of such a modeling effort to *use* when we have accommodated them within a theoretical framework.

2.5.2 Transcriptomics

A new research pipeline involving the application of multiple dimensionality reduction transformations in sequence to transcriptomics data has become popular in recent years. These methods take datasets originally expressing hundreds of thousands of dimensions and successively transpose them down to lower and lower manifolds, ultimately outputting a colorful 2-dimensional plot which researchers interpret both visually and via quantitative metrics. These plots are taken to represent meaningful groupings of samples according to gene-expression. Computational biologists Tara Chiari and Lior Pachter investigated the credentials of these methods, finding them to fall short of good scientific and statistical practice in key respects (Chari and Pachter, 2021). These dimensionality reduction methods are motivated over more traditional techniques in transcriptomics by appeal to their data-driven nature and their relative lack of human input. As the critical review by Chari and Pachter (2021) reveal, this way of selling the methods and their capabilities obscures 1. How they distort or discard data and the informative patterns it is intended to reveal, 2. Aspects of the methods under a researcher's control, 3. The human role in interpreting (or misinterpreting) exploratory visualizations of this nature.

Single-cell transcriptomics offers an approach to inferring cellular-level gene expression. The technique is utilized for identifying cell populations, modeling transcription dynamics, inferring the developmental trajectories of cellular populations, and monitoring changes in cell populations relative to health status. Single-cell transcriptomics emerged with the availability of mass quantities of high throughput RNA sequencing and expression data. It is typical in such exercises to be working with datasets which possess hundreds of thousands of feature dimensions; expression data for thousands of genes across millions of cells (Kobak and Berens, 2019). For this reason, researchers typically employ dimensionality reduction techniques. Dimensionality reduction is a method of mapping a high-dimensional dataset to a lower-dimensional space—or *embedding* higher-dimensional data within a lower-dimensional *embedding space*. Dimensionality reduction techniques are used to distill essential patterns from large datasets, make analyses tractable, and isolate signal from noise. Dimensionality reduction is a method of unsupervized ML, meaning it works by extracting the contours of data, rather than working to extrapolate predefined success criteria encoded in labeled data as in supervised learning.

A now established workflow in single-cell transcriptomics involves applying dimensionality reduction techniques sequentially to high-throughput RNA expression data; first linear methods which reduce the dataset to tens of dimensions using principle component analysis (PCA) or analogous techniques of dimensionality reduction, followed by one of two purpose-built two-dimensional nonlinear reductions: t-distributed stochastic neighbor embedding (t-SNE) and Uniform Manifold Approximation and Projection (UMAP). These methods produce visualizations for both qualitative and quantitative exploratory data analysis.

The intuition behind using dimensionality reduction techniques in this way is as follows. The data on which analyses are run has a certain number of features, which determine the dimensionality of the dataset. The data we are looking at can then be thought of as points lying within a space of that many (n-)dimensions. The specific datapoints we are seeking to characterize lie along a (likewise n-dimensional) manifold within that space. Advanced statistical methods like t-SNE and UMAP are designed to be capable of transposing very high-D data down to low-D embedding spaces while preserving informative structure and discarding irrelevant dimensionality. These methods try to distill the relationships between datapoints and their neighbors and preserve these in projecting the data down to lower-D spaces.

In the transcriptomics workflow under critique, PCA is first run on the original data, creating a linear transformation down to a space with tens of dimensions. This is followed by one of two nonlinear reductions: t-SNE or UMAP. The t-SNE method works within the high-D embedding to calculate pairwise similarities using a t-distribution and then runs a distance-preserving embedding into a lower-D space. UMAP works according to the same general principle, but has more advanced mathematical underpinnings and stronger theoretical guarantees. UMAP first builds what is called a fuzzy simplicial complex and from this constructs a low-D graph optimized to express the salient distances represented therein. UMAP outperforms t-SNE in preservation of global structure.

In order to be useful tools for the kinds of inferences transcriptomics aims for, these methods must faithfully preserve both local and global structure. It can be difficult to independently verify the success of such methods when used in exploratory data analyses with complex data, where ground truth is unavailable. One available means of validating these methods is to draw a very simple shape in high-D, replicate the dimensionality reduction workflow, and determine whether the original structure can be recovered. In their 2021 probe of this now-standard transcriptomics workflow, Chari and Pachter do just this: using UMAP to produce a 2-dimensional representation of a manifold whose structure is already known. The researchers found that not only was the original structure so obscured as to not be inferrable from the resultant embedding, but erroneous groupings of the original structure were introduced in the transformation.

This procedure was one among a series of analyses carried out by Chari and Pachter (2021). Taken together, their assessments demonstrated that the practice of repeated application of dimensionality reduction techniques introduced heavy distortions to the original manifold representation. Critically, Chari and Pachter's analyses reveal that the now-standardized PCA plus t-SNE or UMAP workflow is incapable of preserving the interpretively salient features of the datasets under investigation: local structure, global structure, distance, and continuousness (Chari and Pachter, 2021). What is more, interpretive practices surrounding the resulting visualizations led to erroneous or conflicting conclusions.

Exploratory, data-driven methods like the transcriptomics workflow under scrutiny are motivated on the grounds of their supposed empiricism; they are claimed to be untainted by human bias and unconstrained by existing hypotheses. These applications of dimensionality reduction techniques, in particular, are lauded as handling "all data and all relationships;" however, this common practice "distorts data in obscure ways, attempts to pack the capabilities of many different analyses into one space, and is easily manipulated" (Chari and Pachter, 2021, p.15). Chari and Pachter (2021) refer to the use of dimensionality reduction techniques in transcriptomics as a "blind application" of "heuristic procedures" (p.13), arguing that "there is little theoretical support for this practice" (p.1). As substitute for such (supposed) atheoretical practices, Chari and Pachter endorse "targeted analyses" and "hypothesis-driven biological discovery" (p.15).

While the heart of Chari and Pachter's critique is a demonstration of poor practice involving the combined t-SNE/UMAP workflow, they point clearly to alternative approaches, including semi-supervized learning methods and targeted embeddings for specific featural dimensions. The ambitions of such analyses are, necessarily, more constrained than those of sequential embedding pipelines. They require specifying in advance what features of the data are under investigation. They also require making explicit what assumptions go into the analysis: "it is possible to construct embedding spaces which more explicitly control and improve nearest-neighbor structure and retention," write Chari and Pachter, "[h]owever, such optimizations require making an assumption regarding the appropriate distance/similarity metric, as is generally the case with the neighborhood-based analysis methods ubiquitous across the tasks [on which t-SNE/UMAP are applied]" (p. 14).

The authors of the critique endorse the methods of dimensionality reduction in more limited, principled, and theory-informed applications to transcriptomics: "By targeting the objective of an embedding...one can take advantage of prior knowledge/annotations and more directly determine the necessary dimensionality for a given question" (Chari and Pachter, 2021, 15). Such alternatives require incorporation of domain expertise, critical thinking, and the ability to both identify and (statistically) articulate what you are looking for—

characteristics markedly absent from the t-SNE/UMAP workflows under scrutiny.

2.5.3 Takeaways for ML in scientific practice

The vast majority of applications of the tools of ML to science have been run of the mill: automating laborious processes, achieving minor gains in efficiency or accuracy over human classification or "analogue" statistical techniques without notable breakthroughs in the variety of knowledge gained by their use. The accomplishments of AlphaFold 2.0 are a striking departure from these more quotidian uses. The scientific community has acknowledged AlphaFold as a resounding success; perhaps the greatest win for ML in science to date, if the Nobel Prize is any measure. No other application of ML to science has achieved quite so stark an advantage over pre-existing techniques.

Paralleling the mounting successes of ML in scientific application are a growing number of instances of scientific ML gone wrong: ML-involving research practices deemed deficient by the scientific communities in which they are embedded, e.g., (Goddard et al., 2018; Andrews et al., 2024; Bowers et al., 2023). Some of these replicate the same pattern of errors observed in our transcriptomics case study, and can be traced to the same root cause.

Goddard et al. (2018), for example, offer a critique of a strikingly similar misuse of dimensionality reduction techniques in neuroscience. These are, like the use of PCA, t-SNE, and UMAP in transcriptomics, unsupervised, exploratory uses of dimensionality reduction techniques. Like the transcriptomics case, these are ML methods typically reserved for rote data transformation that have been ambitiously repurposed for inference to the structure of the target system; in this case, not gene expression dynamics, but the neural representation of visual features. Goddard et al. (2018) reveal the inadequacies of these methods by an approach similar to that employed by Chari and Pachter (2021): applying the methods to areas with known ground truth, revealing, in the process, that the methods are incapable of recovering the most basic features of encoding.

The methods of ML are increasingly saddled with more and more ambitious tasks in science: extending far beyond mere signal processing to playing a fundamental role in inferring the structure of our natural world. When ambitious scientific applications of ML, like AlphaFold, have succeeded, it is in virtue of the conceptual resources they have incorporated. In this sense, the conclusion I reach aligns with reasoning expressed in Boge, Srećković et al., and Boon: theory-involvement is a requisite feature of our conceptual instruments in science for them to be capable of elucidating previously unknown truths of our natural world from data. The issue is that the perfectly theory-free vision of ML in science, the primary object of these scholars' concerns, is neither the normal nor necessary operation of these methods. It singles out either a strawman or a failure case.

2.6 Conclusion

It has been alleged by representatives from science and engineering communities as well as philosophers of science—reflected in the texts handled in this chapter—that the methods of ML, loosed on sufficient data, are capable of discovering meaningful patterns, natural joints, or mind-independent truths of their own accord. This is believed to be possible in the absence of input from human theorizing or conceptualization of the target system. Inductive inference, however, is understood by philosophers to rest essentially on theory. A dilemma emerges, forcing us to elect between a refutation of the thesis of theory-freedom or a revision to our standing conception of induction. As I have argued in this text, the ideal of theory-free learning via ML from "raw data" is a confused one. Incorporation of domain expertise is crucial for epistemically responsible deployments of ML, within and without science proper.

Advancing the state of the discourse away from false dichotomies and misdirected concerns is essential, for there is both much that is interesting and potentially novel about ML—DL in particular—and much at stake in its appropriate use. Where to localize theoretical considerations in DL-based scientific workflows appear to differ substantively, along various dimensions, from various canonical modes of scientific or statistical modeling.

On a conventional view of experimental science—one held widely by modern scientists across disciplines—we are typically formulating hypotheses and going out to collect data capable of adjudicating between our hypotheses. Thus the ways in which our conceptual grasp on the target phenomena come into play in how the data represent the target are specific to the epistemic concerns of a particular scientific or modeling exercise. In applied ML, we are often handed data corpora or else construct them from amalgamations of pre-existing datasets. This means that a significant amount of the interpretive work—the work of mapping the data onto target phenomena, imbuing it with representational status and content—is work done before the modeler ever comes in contact with the data. This practice seems to defy Bogen and Woodward's (1988) claim that data are intrinsically limited to serving an evidentiary role in a particular experimental context (Bogen and Woodward, 1988).

Theoretical or interpretive work typically comes in again in the problem formulation, in the engineering or choice of model architecture, and in model training regimes, including choice of hyperparameters and loss or cost functions, as seen in the case of AlphaFold. Theoretical considerations further come in at the level of model evaluation, in our formal assessments of the success of the modeling exercise. Finally, such considerations come into play in what we take ourselves to have learned from the model output and, effectively, in *how the model is wielded*, or the interventions predicated thereon. Undoubtedly, the accelerating adoption of ML-based methods will bring about changes to on-the-ground research practices, including changes to the loci of theoretical input thereon. Such changes, however, will have to be not only domain-specific, but specific to the role with which ML methods are saddled in particular applications.

The landscape of science is also undergoing significant changes today, which are worthy of philosophical scrutiny in their own right. Changes to the social, institutional, governmental, and economic infrastructures that support science, and to the knowledge economies it results in, are a rich philosophical subject. These include the fragmentation and specialization of science, the proceduralization of science, its automation, the progressive increase in the distribution of intellectual labor it involves, the extraction of the knowledge of domain experts and its mechanization and codification into operational formulae. Reactions to the adoption of ML in science have largely framed ML as catalyst to these changes. I wish to counter that we can instead view ML as symptomatic of a much older and deeper trend in the development of scientific practice, one which often replicates the form of the society in which scientific practice is embedded in its social structure, its economic model, and its governance. The causal arrow, therefore, may run as much from the automation of scientific practice and the balkanization of scientific expertise to the adoption of the tools of ML as it does in the reverse.

CHAPTER 3

MODELS OF DATA & MODELS OF THEORY: A DISTINCTION WITHOUT A DIFFERENCE?

3.1 Introduction

A bedrock conceptual tool within modern philosophy of science is the distinction that separates models of theory from models of data. The initial account of these modeling forms, due to Suppes (1962), was developed according to the semantic view of theories and drew on attendant accounts of scientific representation. With the so-called "pragmatic turn," however, philosophers of science have rejected both the logical and representational commitments of the prior paradigm on which the original distinction hung. The distinction, as originally conceptualized, appears to collapse. Yet it seems that scientific models nevertheless differ qualitatively in the epistemic status of their empirical content: some formal models express conjectural content, while others express content which has been validated by observation or measurement. The original distinction between models of theory and models of data partially tracks such a difference. However, in setting out a hard and fast distinction between species of conceptual tool where no such systematic difference can be found in scientific practice, the Suppesian account of models of data and models of theory threatens to obscure important differences which exist in the empirical content of scientific models. If philosophers of science are to have a distinction between models of data and models of theory, it must be redrawn; gradations of difference in the degree of confirmation of empirical content are a strong contender for regrounding this modeling taxonomy.

Scientific inference is generally understood as a procedure of putting theory into contact with the results of observation or measurement—in other words, data. Philosophers of science characterize this contact as facilitated by models. Suppes (1962b) proposed, more precisely, that this contact occurs by means of a theoretical model being put into contact with what he called a model of the data, or data model. On this conceptualization of science, theoretical models represent theoretical constructs. A model of data, meanwhile, is taken to be a direct transformation of the prepared data that emerges from an observational or experimental regime.

What is made clear within Suppes' 1962 proposed account of the data model is that it is a distinct species of logical entity from theories and theoretical models. The role with which the data model is saddled in scientific practice also distinguishes it from theoretical models: data models are intended to be confirmatory of the theoretical posits embodied in theoretical models. Suppes' original account was notoriously brief, leaving core features of the account unarticulated—and leaving the literature in philosophy of science to fill in the blanks. These included details about how data models stand in relation to target phenomena in nature, their representational status, and how this differs from theoretical models.

While different paradigms in philosophy of science have arrived at diverging accounts of what a theoretical model consists in, the notion of a data model has persisted relatively unchanged and unchallenged through successive waves of philosophical description. The modern tradition of philosophy of science has cycled through at least three foundational paradigms over the course of its history, including the syntactic view, the semantic view, and the pragmatic view (Winther, 2015). The distinction in question emerged under the semantic view, which was the first paradigm to emphasize an essential role for the scientific model. Contemporaries of semantic theorists advocated for a view referred to as "models-as-mediators" (Morgan and Morrison, 1999). Many of the ways in which this account diverged from the semantic view were inherited by the subsequent pragmatic view. The division between data models and theoretical models, however, has permeated each of these paradigms in turn and without explicit adjustment.

This is striking, as subsequent paradigms appear to reject most of the core principles on which the distinction was initially based. These include such core matters as the ontology of what models are taken to consist in, their logic reconstruction, their representational status, how data is understood, and how responsive these philosophical accounts are to the details of scientific practice. In the first place, these various paradigms diverge on what entities in scientific practice they pick out as "models" or "modeling." On the semantic view of theories, the notion of a "model" refers roughly to the representational use of mathematics in science. These mathematical models as wielded by scientists are understood by semantic theorists as models in a model-theoretic sense.[1] The models-as-mediators and pragmatic views accept a much more expansive conception of models, inclusive not only of mathematics but physical models, scale models, diagrams, maps, model organisms, and the like. The semantic view is also set apart from subsequent accounts in its commitment to a logical or metamathematical reconstruction of the conceptual tools of science. On the semantic view, theories are collections of logical sentences, with theoretical models standardly understoond as set-theoretic entities instantiating values for the theoretical constructs and preserving the relationships described therein (Suppes, 1962b). Subsequent accounts abandon the logical and set-theoretic commitments of the semantic view (Bokulich and Parker, 2021; Winther, 2015).

Although there is considerable diversity within each of the philosophical paradigms here mentioned in their views on representation, the semantic view initially came packaged with a family of views on representation that all share in taking representation to be a mind-independent, two-place relation (Knuuttila, 2005; Suárez, 1999; de Oliveira, 2021b). Subsequent accounts distinguish themselves in agent and goal-relativizing their conceptions of representation. As has been pointed out by several interlocutors in the modeling conversation, the semantic view of theories largely held data to be "raw" and "worldly," sometimes even identifying it directly with target phenomena (Bokulich and Parker, 2021; Harris, 2003; Leonelli, 2019b). The Suppesian notion of data model rests on such a conception of data.

1. Whether this is some deep ontological commitment, a rational reconstruction, or merely a convenient gloss is in general not specified in the literature and might vary between interlocutors.

Finally, earlier waves in philosophy of science were more a prioristic in their reasoning, and were quick to generalize over science at large from rational reconstructions of particular practices. The models-as-mediators and pragmatic accounts emerged consonant with the "practice turn" in philosophy of science. These paradigms proceed from attentive study of research operations *in situ*, acknowledging the complexity and diversity of these practices.

The notion of data models as categorically distinct from models of theory is grounded in specific commitments of the semantic view of theories and attendant accounts of representation: ontological, logical, representational, and practical. I argue that shifting views on the logical and representational status of the two genres of model under scrutiny, in particular, render the distinction incompatible with commitments of the pragmatic turn. This argument extends beyond existing critiques of the distinction which note that the concept of data model appears to hang, essentially, on a characterization of data as "raw"—at a stark disconnect to treatments of scientific data under the pragmatic view (Bokulich and Parker, 2021; Harris, 2003; Leonelli, 2019b).

The distinction does, however, appear to track—if only loosely—something crucial to our understanding of science and its employment of models: that scientific models differ in the extent to which they have received empirical confirmation. In fact, it may be ventured that the (apparent) ability to track this property of models accounts for the persistence of the distinction in question. I argue, however, that the existing distinction between models of data and models of theory does not facilitate this capability, but rather hinders it, creating differences where there are none and obscuring differences that matter. Models that look canonically like data models can often express conjectural content; meanwhile, models that look definitionally like theoretical models can encode empirical content that has received extensive confirmation. Often one and the same model represents a mix of empirical contents which are heterogeneous in extent of verification. Philosophy of science will indeed require tools within its conceptual arsenal capable of capturing differences in the status of empirical content. The distinction between models of data and models of theory, however, cannot be relied upon to do this work—at least not in its original incarnation. If the distinction is to be rehabilitated, it should be redrawn so as to track distinctions that matter for the nuances of actual modeling practices.

My argument proceeds as follows. Section 2 introduces Suppes' initial (1962) account of models of data, focusing on his conception of what features distinguish them from models of theory. This section then looks to the reception of this conceptual distinction in the philosophy of science literature for a fuller account of how the distinction is used and understood. I next provide evidence that the distinction persists within contemporary, 'pragmatic' accounts of scientific modeling. Section 3 describes the philosophical paradigm under which the conception of data models emerged: the semantic view, and associated views on scientific representation. I look at some of the essential tenets of this paradigm and describe how the subsequent paradigm, dubbed a pragmatic view, has distanced itself from these commitments. On these grounds, I make a case for the incompatibility of the original distinction with the commitments of pragmatic views. A feature of modeling that Suppes' distinction partially tracks is the extent to which the contents of a model has received empirical confirmation. In Section 4, I make the case that the existing theory model/data model distinction cannot effectively pick out differences in the status of empirical content. I urge that the literature should abandon the existing taxonomy in species of scientific model to the development of tools capable of parsing the epistemic status of a model's content.

A terminological note before I begin: Part of my aim in this chapter is a targeted assessment of what differentiates competing philosophical accounts of models and their representational status. This direct comparison is made impossible if the various paradigms under comparison are referring to entirely different things with the term "model." The semanticists—we have established—tend to think of theoretical models as roughly synonymous with representational use of mathematics in science. Philosophers of science within the models-as-mediators and pragmatic paradigms, on the other hand, have a much more pluralistic conception of what counts as a scientific model. Not only mathematical systems, but physical systems, model organisms, and graphical representations, to name merely a few examples, are all characterized as models within the pragmatic paradigm. Because what the semantic account views as models is a proper subset of what the models as mediators and pragmatic accounts conceive of as models, I will go with this understanding of what constitutes a scientific model. An additional reason to proceed with this definition is the a priori implausibility that one unified account could make sense of the representational status and epistemic purchase of things as diverse as laboratory mice, miniature cities, and systems of differential equations. I will, therefore, restrict the discussion of models to representational uses of mathematics in science.

3.2 The Distinction

3.2.1 Initial Presentation

Working within this semantic paradigm, Suppes (1962b) introduced a distinction between models of data and models of theory. According to this original account, data models and theoretical models play non-overlapping roles in scientific inference, and are distinct in logical (metamathematical or set-theoretic) structure. Suppes motivates the introduction of the data model concept as necessary to account for steps in the process of bringing theory into contact with empirical results: "exact analysis of the relation between empirical theories and relevant data calls for a hierarchy of models of different logical type. Generally speaking, in pure mathematics the comparison of models involves comparison of two models of the same logical type, as in the assertion of representation theorems. A radically different situation often obtains in the comparison of theory and experiment" (Suppes, 1962b, p.253). Models of data are defined on Suppes' account in contrast to theoretical models, of which by this point there existed a relative consensus in the philosophy of science literature.[2] Suppes inherits his conception of a model from Tarski's logical system adopting the Tarskian definition of a model as: "[a] possible realization in which all valid sentences of a theory T are satisfied" (Tarski, 1953, p.11).

Suppes summarizes the going account of theoretical models as follows: "a model of a theory may be defined as a possible realization in which all valid sentences of the theory are satisfied, and a possible realization of the theory is an entity of the appropriate set-theoretical structure" (Suppes, 1962b, p.252). By contrast, models of the data are defined "in terms of possible realizations of the data. As should be apparent, from a logical standpoint possible realizations of data are defined in just the same way as possible realizations of the theory being tested by the experiment from which the data come. The precise definition of models of the data for any given experiment requires that there be a theory of the data in the sense of the experimental procedure, as well as in the ordinary sense of the empirical theory of the phenomena being studied" (Suppes, 1962b, p.253).

The original (1962) manuscript in which Suppes introduced the concept of data model was brief and lacking in much detail. A subsequent literature in philosophy of science has received the distinction and offered much clarification and elaboration. Frigg and Nguyen (2017a) Explicate the concept as follows:

> "What are data models? Data are what we gather in experiments. When observing the motion of the moon, for instance, we choose a coordinate system and observe the position of the moon in this coordinate system at consecutive instants of time. We then write down these observations. The data thus gathered are called the raw data. The raw data then undergo a process of cleansing, rectification and regimentation: we throw away data points that are obviously

2. Largely because the philosophers of science had only just begun to treat the subject matter of modeling practice; the first few scholars to write on the subject—Suppes, Suppe, van Fraassen, Giere—took after Suppes' initial proposition.

> faulty, take into consideration what the measurement errors are, take averages, and usually idealise the data, for instance by replacing discrete data points by a continuous function. Often, although not always, the result is a smooth curve through the data points that satisfies certain theoretical desiderata" (Frigg and Nguyen, 2017a)

A passage in Van Fraassen (2008) illustrates the concept well:

> "[O]n the weather forecast website I consult I can find a graph depicting yesterday's temperature plotted against time. This was constructed from data gathered at various stations in the region, at various times during the day—this graph is a smoothed-out summary of the information that emerged from all these data, it is a *data model*. The question about the daytime temperatures in this region of one day ago is answered with a *measurement outcome*, certainly—but that is the graph in question, which is a data model constructed from an analysis of the raw data."(Van Fraassen, 2008, p.166)

To get a sense of the modern consensus account of this distinction in the field, it will be useful to appeal to publicly available introductory resources, such as the *Stanford Encyclopedia of Philosophy*. In their *Stanford Encyclopedia of Philosophy* entry, Frigg and Hartmann (2012 (archived version) describe the data model concept in the following terms:

> "A model of data (sometimes also 'data model') is a corrected, rectified, regimented, and in many instances idealized version of the data we gain from immediate observation, the so-called raw data (Suppes 1962). Characteristically, one first eliminates errors (e.g., removes points from the record that are due to faulty observation) and then presents the data in a 'neat' way, for instance by drawing a smooth curve through a set of points. These two steps are commonly referred to as 'data reduction' and 'curve fitting'. When we investigate, for instance, the

trajectory of a certain planet, we first eliminate points that are fallacious from the observation records and then fit a smooth curve to the remaining ones. Models of data play a crucial role in confirming theories because it is the model of data, and not the often messy and complex raw data, that theories are tested against." (Frigg and Hartmann, 2012 (archived version, Webpage)

The basic account of the distinction between theoretical models and data models holds that models of theory are set-theoretic structures instantiating values consistent with the stipulations of a theory, while data models are mathematical transformations of raw data. The models are introduced as distinct in their provenance: one is derived from theoretical resources, the other a manipulation of experimental data. These models are further distinguished from one another in their logical formulation. Crucial details of the account, however, are not made explicit within Suppes' original exposition. The most important of these concern the representational status and contents of these two genres of model. Are both to be understood representationally? If so, do models of theory and models of data share the same targets of representation? Is the relationship between representational vehicle and target the same in each instance?

3.2.2 Augmentations & Adaptations of the Distinction

Among those who have weighed-in on the representational status and contents of data models there is quite a bit of variance in the details. All philosophers who have addressed this matter in print, however, seem to agree that Suppes intended for data models and theoretical models to diverge along the lines of what, and or how, they represent.

In their overview of the philosophical literature on *Models in Science*, Frigg and Hartmann (2012) summarize the received view as follows: "[m]odels can perform two fundamentally different representational functions. On the one hand, a model can be a representation of a selected part of the world (the 'target system'). Depending on the nature of the target, such models are either models of phenomena or models of data. On the other hand, a model can represent a theory" (Frigg and Hartmann, 2012 (archived version). Reutlinger et al. (2018) clarify that toy models, a variety of theoretical model "refer to a target phenomenon...as opposed to...models of data" (p.1070).

Van Fraassen maintains that models of theory represent models of data (or, in van Fraassen's idiom, "empirical substructure"). The *Internet Encyclopedia of Philosophy* summarizes the position as follows:

> "The targets of theoretical models on the semantic view are not always real world systems. On some views, there is at least one other set of models which serve as the targets for theoretical models. These are variably called empirical substructures (van Fraassen 1980) or data models. These are ways of structuring the empirical data, typically with some mathematical or algebraic method. When scientists gather and describe empirical data, they tend to think of and describe it in an already partially structured way. Part of this structure is the result of the way in which scientists measure the phenomena while being particularly attentive to certain features (and ignoring or downplaying others). Another part of this structuring is due to the patterns seen in the data which are in need of explanation. On some views, most notably van Fraassen's (1980), the empirical model is the phenomenon which is being represented. That is to say, there is no further representational relationship holding between data models and the world, at least as scientific practice is concerned" (Boesch, 2015).

For van Fraassen, data models, or empirical substructures, *just are* the natural phenomenon under study, not representations of the phenomena. "The whole point of having theoretical models," Van Fraassen writes, "is that they should fit the phenomena, that is, fit the models of data" (Van Fraassen, 1980a, p.667). Boesch (2015) highlights that for van Fraassen, the data model just "is the phenomenon." Brading and Landry (2006) write:

"van Fraassen...suggests that we simply identify the phenomena with the data models: 'the data model . . . is, as it were, a secondary phenomenon created in the laboratory that becomes the primary phenomenon to be saved by the theory' (van Fraassen 2002, 252). In this way, the step from presentation to representation is made almost trivially: the data models act as the 'phenomena to be saved' Brading and Landry (2006, 578). In summary, "van Fraassen...simply asserts the identity of the data models and the phenomena" Brading and Landry (2006, 580). On van Fraassen's account, models of theory are then taken to represent data models in virtue of an isomorphism holding between the two structures. In contrast, Harris (2003) takes models of data to represent worldly phenomena: "there are different ways to model data and...these different ways of modeling result in data models that have differing known similarities and dissimilarities with the real world objects they are intended to represent" (Harris, 2003, p.7). The representation relation Harris takes to hold between model and target appears to consist in similarity or shared features. Isomorphism and similarity are both classical, two-place "mapping" views.

The texts here surveyed serve to paint a picture of the reception of Suppes' account of data models in the philosophy of science literature. They offer a disjointed account of what models of theory and models of data are intended to represent, and what that representation relation should be taken to consist in. These various scholars are unified, however, in having reached the conclusion that Suppes intended for data models to be understood as distinct from models of theory in either representational status, representational content, or both. The reception of the distinction in the literature has likely inferred representational differences on the basis of the distinct epistemic roles Suppes laid out for these modeling types. If models of theory and models of data are intended to serve non-overlapping epistemic roles in scientific inference, they cannot be identical in representational status and content, according to the commitments of the semantic view. It has been highlighted as characteristic of this family of views that they wholesale reduce the epistemic purchase of models to their representational function (Batterman, 2010; Grüne-Yanoff, 2013; de Oliveira, 2021b). If data models and theoretical models are intended to play distinct roles in licensing empirical inference—as Suppes intended—they cannot represent the same content in the same manner.

3.2.3 Persistence of the Distinction Past the Pragmatic Turn

Accounts of the distinction in Suppes (1962b), Suppes (2002), and (Van Fraassen, 1980b) are operating squarely within the semantic view of theories. That recent surveys of the state of play in the philosophical literature on models and representation—including Frigg and Hartmann (2025 (latest version) and Boesch (2015)—speak of data models among the canonical modeling types attests to the perception of the continued relevance of this distinction. I turn in this subsection to evidence that the distinction between models of data and models of theory is taken on board by philosophers working within the pragmatic paradigm in philosophy of science.

Harris (2003) endorses a similarity account of representation, a type of mapping account, and is therefore consistent with the representational commitments of the semantic view. In other respects, however, Harris challenges the received distinction, in particular, in its commitment to the notion that the data produced in experiment is "raw." What emerges from an experiment, Harris argues, is often not "raw data," but a fully-fledged and already interpreted data model. Harris writes "a data model" ...is not simply a copy of what happens. Instead it incorporates elements of theory to create a representation that contains features of interest to the scientist" (Harris, 2003, p.16)

Like Harris (2003), Bokulich and Parker (2021) and Leonelli (2019b) push back against features of the received account of the distinction, but leave it fundamentally intact. Bokulich and Parker (2021) "critically examine two common ways of thinking about data and data modeling—the mirroring view and the set theoretic view—and motivate the need for an alternative understanding of data" (p.31). They argue against the logical or set-theoretic ontological grounding of the Suppesian hierarchy of models. They also argue against features of the semantic view's representational commitments: urging a pragmatic view on representation over mapping accounts.

Intriguingly, Bokulich and Parker (2021) critique both Giere (1999) and Leonelli (2016) on the grounds that both authors expose severe limitations with the semantic or Suppesian account of models, all the while holding onto the Suppesian notion of data model, failing to recognize how their own critiques extend to the concept. "Giere's critique" Bokulich and Parker (2021) write, "centers on the instantial view of theoretical models. Yet Suppes adopts the same instantial view when he speaks of 'models of data'...Giere, in the same (1999) paper that challenges Suppes' instantial account of theoretical models, has a section on 'Models and Data' where he endorses Suppes' notions of models of data, and related hierarchy, without extending his critique" (Bokulich and Parker, 2021, p.4-5). "Surprisingly," they continue, "this aspect of Suppes' view has gone unchallenged (or perhaps not fully recognized) in many subsequent discussions of his views on data models" (Bokulich and Parker, 2021, p.4).

Bokulich and Parker (2021) "pragmatic-representational" revision to the Suppesian depicting of data modeling works in the direction of compatibilizing the distinction with the pragmatic turn. I believe, however, that the authors do not recognize the force of their own critique. In abandoning the ontological and representational commitments of the semantic view, the distinction between data model and theoretical model is left without a leg to stand on. It is striking that Bokulich and Parker (2021) chastise their peers for holding onto a distinction which appears to clash with their theoretical commitments, while nevertheless holding onto the self-same distinction in spite of its apparent conflicts with pragmatic views.

Leonelli (2016; 2019) is placed in a similar boat to Bokulich and Parker (2021). Bokulich and Parker (2021) find Leonelli culpable of "[t]he same elision" as Giere in her (2016) book, though they acknowledge that her (2016) treatment of data models does much to revise the view and compatibilize it with her (pragmatic) views on representation and data. Nevertheless, Leonelli upholds the fundamental distinction between models of data and models of theory (Leonelli, 2016, 2019b).

Antoniou (2021) endorses the distinction between models of theory and models of data as discrete stages in a scientific inference and saddled with discrete roles. Antoniou applies the account to an experimental regime at the CERN Large Hadron Collider (LHC) for testing the lepton flavour universality of B-meson decays. Antoniou (2021) aligns himself with the pragmatic view, endorsing Bokulich and Parker (2021)'s modifications to the Suppesian account of data models. Antoniou defines the notion of data model as "the representation of an experimental result in the form of a graph, table or numerical answer that allows the comparison of experiment with theory" (Antoniou, 2021, p.30). Antoniou notes that "[w]hile this straightforward answer to the question of what a data model is does not differ from what Suppes and others have said, what is of special philosophical interest is the complicated and extremely laborious process of constructing a data model in [High-Energy Physics], which has largely been overlooked by philosophers of science" (Antoniou, 2021, p.30).

Taken together, these amassed contributions from philosophers of science constitute evidence that the distinction between models of theory and models of data persists within the contemporary literature in philosophy of science which associates itself with the "pragmatic turn." While the Suppesian notion of these distinct species of modeling activity has been subject to scrutiny and modification by philosophers operating with the resources of this new paradigm, the essence of the distinction has remained unscathed. The notions of data model and theoretical model remain actively in use within philosophical analyses of scientific practice.

3.3 Changing Paradigms

The tradition of philosophy of science that emerged in the twentieth century is classically reconstructed as having undergone three waves in its understanding of the conceptual tools of science. These paradigms are named after targets of linguistic analysis: syntax, semantics, and pragmatics. The first of these, dubbed the syntactic view, was associated with the work of the logical empiricists. This paradigm conceived of theories as collections of sentences in a predicate logic; the primary work of philosophy of science was then relegated to the axiomatic reconstruction of these theories. The relation between theory and natural phenomena was facilitated by observation sentences in an observation language.

The semantic view of scientific theories diverges; while theories are still reconstructed as collections of sentences, the semantic view outlines an indispensable role for scientific models. Its foremost proponents include Van Fraassen (1980b), Giere (1988), Suppe (1989), and Suppes (1962b). Suppes, a progenitor of the semantic view, accounts for models as set theoretic structures instantiating possible values for the constructs specified thereunder and satisfying all valid sentences therein. Van Fraassen, likewise a pioneer of the semantic view, diverges somewhat in his ontological commitments. Van Fraassen exchanges Suppes' set theoretic commitments for a view of the nature of models grounded in state space representations.

The pragmatic turn abandons the logical underpinnings of the syntactic and semantic views, analyzing theories and models in terms of their roles assumed in the human practices of science rather than in terms of predicate logic, set-theory, or state space representations. The pragmatic view inherits from the models-as-mediators perspective the view that models are where the primary action of scientific inference takes place, alongside the idea that they act relatively autonomously from theories. Importantly, pragmatic accounts assess the conceptual tools of science primarily in terms of their *use*: what predictions or interventions they facilitate. The epistemic utility of models understood in agent and goal-relative terms: "there has also been a pragmatic turn: many philosophers now emphasize that scientific modeling is an activity undertaken by agents with specific goals and purposes in mind," (Bokulich and Parker, 2021, p.31) For present purposes, we are primarily interested in the commitments of the semantic view of theories, and how pragmatic accounts have rejected these commitments and supplanted them with their own. In particular, we are interested in the dimensions of logical reconstruction (or lack thereof), how representations are understood, and commitments (or lack thereof) to the nature of data, so I will divide up the discussion along these axes.

3.3.1 Logical Structure

The syntactic and semantic accounts shared in assessing the conceptual instruments of science—theories and models—in logical, meta-mathematical, or mathematical terms. Especially on the semantic view of theories, there is some heterogeneity and general fuzziness concerning whether this portrayal of theories and models in formal language represents deep ontological commitment or merely a useful rational reconstruction. As we have covered, Suppes (1962; 2002) conceived of models of theory and models of data as of a fundamentally "different logical type." If the logical (or set theoretical) distinctness of these two types of model is to do real work, however, it must be a difference in more than mere convenience of description. We must therefore grant that distinctness of logical type some ontological bite if we are to grant that the reality of the distinction in modeling types hangs (at least in part) thereon. With the pragmatic turn, however, came a wholesale abandonment of the formal commitments of the prior paradigms. Straightforwardly, then, in abandoning the formal commitments of the semantic view, what distinguishes theoretical models from data models within the pragmatic paradigm cannot be a difference of "logical type."

3.3.2 Qualities of Data

Philosophy of science prior to the pragmatic turn had yet to articulate an account of data, and it was standard fare for philosophers to offer reconstructions of scientific inference in which data were presumed to offer a kind of direct and unmediated access to reality (Leonelli, 2016). In fact, some philosophers simply identified data with phenomena in nature, prompting

Bogen and Woodward's (1988) critical injunction, which urged the necessity of maintaining a conception of phenomena discrete from data (Van Fraassen, 1980b; Bogen and Woodward, 1988). One of the essential developments of the pragmatic turn has been an articulation of an account of the role of data in science: what constitutes data, how it relates to target systems, how it undergoes 'journeys' from collection to processing and archiving before finally facilitating inference (Leonelli, 2016). Proponents of the pragmatic view dispense with the idea that there can be such a thing as 'raw' data which can be reasonably identified with worldly phenomena or which provide direct access to it. Under the pragmatic view, data are taken to be a component of scientific practice whose epistemic purchase over nature must be explained in their own right.

Suppes' (1962, 2002) account puts forward the notion of data model as a means of bridging the divide between our scientific theories and "raw experimental experience," or "raw data." The reliance of the distinction between models of data and models of theory on a conception of data as "raw" presents two challenges. In the first place, as has been noted elsewhere, that the Suppesian account bottoms out at the level of data means it stops short of accounting for how models (and, therefor, science) enable us to learn about the real world.

As Leonelli articulates the problematic: "The contemporary characterisation of data models as 'corrected, rectified, regimented and in many instances idealized version of the data we gain from immediate observation, the so-called raw data' (Frigg and Hartmann 2016) demonstrates how the very distinction between data models and other types of models is predicated upon presupposing the existence of 'raw data' resulting from 'immediate observation' of the world, and thus arguably providing direct and unmediated access to it" (Leonelli, 2019b, p.21-22).

The second issue has a more immediate bearing on our thesis: If the Suppesian notion of data model is defined relative to a conception of data as raw, hence being either identifiable with worldly phenomena or providing direct access to them, what remains of the distinction once we have progressed beyond a conception of data as raw? What revisions to the conception of data model would be necessary to make such an account compatible with the conception of data under the pragmatic view?

On the original articulation of the account, data models are taken to have a kind of special epistemic purchase over natural phenomena, a quality they inherit from raw data in virtue of its supposed 'unmediated' access to reality. If we abandon this conception of data, the epistemic purchase of data and data models must be otherwise accounted for. The standard way of accounting for the epistemic purchase of scientific models is to give an account of their representational capacities—in fact, many take it is definitional of a model that its usefulness is reducible to its representational function (Frigg and Nguyen, 2017a; de Oliveira, 2021b). Under the pragmatic view, with the epistemic purchase of data over nature no longer a given, an account of how data and data models *represent* the world becomes necessary. This, however, presents a further set of problems, as we shall see in the next section.

3.3.3 Views on Representation

The development of philosophical accounts concerning the representational status of theories and models is interwoven with the development of accounts of the structure of scientific theories (and models thereunder). There is a tight link, therefore, between the semantic view and "mapping" accounts of representation (Contessa, 2007)—many philosophers, indeed, treat these positions as synonymous (Suárez, 1999). Suárez, a foremost critic of mapping views of representation, writes: "I argue against an account of scientific representation suggested by the semantic, or structuralist, conception of scientific theories. Proponents of this conception often employ the term 'model' to refer to bare 'structures', which naturally leads them to attempt to characterize the relation between models and reality as a purely structural one" (Suárez, 1999, p.75).

Numerous additional critics of the received view on representation have further established that characteristic of this family of views is a commitment to an account of the epistemic status of models that is fully supervenient on their representational properties (e.g., (Batterman, 2010; Grüne-Yanoff, 2013; de Oliveira, 2021b)). On the semantic-mapping family of views—within which the distinction between models of data and models of theory was conceived—these two classes of model are widely taken to differ in representational and epistemic status (Suppes, 1962b; Frigg and Nguyen, 2017a; Reutlinger et al., 2018). Summarizing the received view, Frigg and Hartmann attest that "models can perform two fundamentally different representational functions" (p.1). Reutlinger et al. (2018) clarify that toy models, a variety of theoretical model "refer to a target phenomenon...as opposed to...models of data" (p.1070).

There exists some diversity (and some inconsistency) within this family of views concerning what, precisely, that difference consists in. Proponents of a semantic–mapping accounts have largely left the matter of how data models, or even data themselves, are understood to represent the world within the semantic–mapping framework unspecified. Suppes himself refers to models of theory as set theoretic structures that instantiate the values of theoretical terms within a theory. Models of data, on Suppes' account, instantiate the values of an experimentally derived dataset. How this dataset, or a model constructed therefrom, in turn represents the world is left unspoken. "The Suppesian construal of data models," write Bokulich and Parker, "leaves the relation between data models and the world at best unanalyzed, and at worst erased" (Bokulich and Parker, 2021, p.31). Indeed, as Leonelli (2019b, 2016) has suggested, the nature of data has been largely overlooked within this philosophical tradition, and it may be that essential features of semantic-mapping accounts bar them from being able to satisfactorily describe the connection between data and worldly phenomena.

Proponents of semantic-mapping style accounts once widely endorsed a view of theoretical models as representing reality by means of resemblance or shared structural features with physical phenomena. Many of these scholars have since clarified this view as a contention that models of theory resemble or share structure with models of data (Brading and Landry, 2006; Frigg and Nguyen, 2017a; Knuuttila, 2005). Such clarifications add credence to the idea that proponents of semantic-mapping views hold data to be, in some sense, "worldly," at a difference to theoretical models. This is evidence by the fact that many proponents of such accounts simply associate data or data models with the phenomena, or even the target system under study, using these phrases somewhat interchangeably (Bokulich and Parker, 2021; Brading and Landry, 2006; Van Fraassen, 1980b). We are further licensed to draw this conclusion because the only alternative available to proponents of such a theory is a kind of retreat to idealism, which scholars under the semantic view explicitly disavow (Brading and Landry, 2006).

Van Fraassen (1980b), for instance, attests that models of data (otherwise known as empirical substructures) *just are* the phenomena of interest, and are represented by theoretical models via isomorphisms obtaining between the two species of model. Considerable push-back against this approach (what he has termed the "loss of reality" objection) has forced van Fraassen to amend his views. In his 2008 treatment of the topic, van Fraassen takes up a position much closer aligned to pragmatic accounts: models of data and models of theory represent in virtue of the involvement of an epistemic agent (Van Fraassen, 2008; de Oliveira, 2021b). This three-place, agent-involved conception of scientific representation is a hallmark of the pragmatic turn, and marks a clear departure from the semantic view and associated 'dyadic' views on representation.

Setting these nuances aside, it is sufficient for the purpose of our argument that the representational status of data models is held to be distinct from that of models of theory on semantic-mapping accounts (Frigg and Nguyen, 2017a; Reutlinger et al., 2018). The philosophical literature on scientific modeling and representation is further committed to the view that "modeling is epistemically valuable because of its representational nature; i.e., the representational relation between model and target is what secures the epistemic worth of modeling" (de Oliveira, 2021b, p.213).

Batterman (2010) observes that what he terms "mapping accounts" "all hold that some kind of accurate mirroring, or mapping, or representation relation between model and target is responsible for the explanatory success of the model" (Batterman, 2010, p.351). Here explanatory success can be identified with epistemic success or utility. Frigg and Nguyen (2017a) attest that "models must be representations: they can instruct us about the nature of reality only if they represent the selected parts or aspects of the world we investigate...if we want to understand how models allow us to learn about the world, we have to come to understand how they represent" (Frigg and Nguyen, 2017a, p.49)

In summary, semantic-mapping accounts hold that the epistemic purchase of models over phenomena in nature is entirely supervenient on their ability to *represent* said phenomena. What this means for our purposes here is that, accepting the tenets of the semantic-mapping family of views, the epistemic purchase of models cannot be different without a difference in representational status or content; conversely, if two models differ in representational status or content, so, too, must their epistemic purchase over phenomena.

Adherents to the semantic view ascribe to a family of philosophical accounts concerning how models represent phenomena, variously dubbed "mapping" (Batterman, 2010; Pincock, 2011), "substantive" (Boesch, 2015) "structuralist" (Suárez, 1999) or "dyadic" (Knuuttila, 2005) accounts. For simplicity and terminological consistency, we will refer to these collectively as "mapping accounts." These mapping accounts all share in specifying epistemic representation as a two-place relation, which holds objectively between representandum and representans. This is variously explicated in terms of similarity, resemblance, shared structure, isomorphism, or partial isomorphism.

There is likewise considerable diversity in accounts of what the epistemic purchase of models over the world is taken to be within the pragmatic view. Some (e.g., Batterman

(2010), Grüne-Yanoff (2013), de Oliveira (2021b) seek to dispense altogether with the idea that the essential epistemic relationship holding between models and the world should be understood representationally. Pragmatic accounts that continue to endorse representation—and these form the majority—are unified in understanding representation to be agent and goal relative (Bokulich and Parker, 2021; Leonelli, 2019b).

What happens when we abandoned the idea that the data, and the data model *just are* the target phenomena in nature which are our objects of study? The only philosopher to have explicitly endorsed such a position is Van Fraassen (1980b), in his earlier works. As we have mentioned, van Fraassen appears to have since (2008) disavowed this version of the account, as sufficient argument against it had mounted as to sink the view (Brading and Landry, 2006; Boesch, 2015; Van Fraassen, 2008). If models of data are not identical to target phenomena and yet have epistemic purchase over target phenomena, then, on semantic-mapping views, it is in virtue of representing said phenomena. But the same must go for models of theory. If some scientific inference over a target phenomenon P is licensed in virtue of putting a model of data into contact with a model of theory, it must therefore be in virtue of both sharing the same target of representation P. Theoretical models and data models therefore cannot be differentiated along lines of *representational content* or *brute representational status*: both must represent, and both must represent the same target phenomenon. Can they differ in their mode of representation? On semantic–mapping accounts, we might draw distinctions along lines of different shared features or distinct structural mappings holding between model and target. On the agent and goal-relative accounts of scientific representation endorsed by the pragmatic view, however, no such distinction can be cleanly drawn. A distinction in modality of representation would have to be assessed in terms of systematic differences in construal or interpretation or human practices surrounding the usage of models in scientific practice. This brings us, naturally, to an analysis of human practice, on which a revised version of the distinction might yet be drawn.

3.3.4 Failure to Generalize

The paradigm under which the distinction between models of data and models of theory was originally drawn rested on a great deal of philosophical baggage, both in terms of its logical and ontological commitments and in terms of its varied and elaborate views on representation. The pragmatic turn differentiates itself in replacing much of this formal infrastructure with careful study of the human practices of which science is composed.

We have established thus far that most vestiges of the initial distinction between models of data and models of theory cannot be maintained under pragmatic accounts. These now outmoded elements include the difference in logical formulation, the reliance on the concept of raw data, the difference in representational content or modality, and the difference in epistemic role, insomuch as we buy into the received opinion that this must be supervenient on representational status. What remains for us to explore is the viability of a distinction between models of data and models of theory grounded in observed differences in human practice.

Suppes' (1962; 2002) account was put forward as a universal account of how scientific inference works; and how it must, necessarily, work. Suppes' initial illustration of the distinction appealed to a case study in mathematical psychology, specifically, learning theory. The case study is illustrative. It also provides clear evidence that, in at least one instance of scientific practice at one point in history, there existed something that looked and acted canonically like a data model and something that looked and acted canonically like a theoretical model. No doubt more such cases could be found, if one is looking. In order for the account to do the work Suppes intended for it, though, it would seem that a distinction between these two modeling types or practices would have to be necessary and universal. Under the pragmatic turn, we are forced to scale back the ambitions of the account. The goal might be revised to providing a sketch of broad contours of variants of model which appear frequently in scientific practice. Even with such tempered ambitions, for us to grant the validity of the distinction, we would need to see a pattern of difference in human practice that is both widespread and systematic.

I argue that no such widespread and systematic a difference can be found, and that the distinction therefore fails to generalize beyond a limited set of simplistic, hypothesis-driven research protocols. As evidence for this claim I will broadly characterize genres of research practices, rather than appealing to case studies, as the target here is a generalization over scientific practice in general, and not a contention over whether or not particular instances of scientific practice conform to the details of certain depictions.

In the first place, scientists today, across nearly all disciplines of science, characterize their work variously as *explanatory* or as *exploratory*(Sheppard, 2020). Setting aside whether such a distinction can be made philosophically intelligible in any deep way, the distinction does seem to pick out an actual systematic difference in scientific practice. Explanatory research, as working scientists describe it, seeks to elucidate causal mechanisms, reveal deeper principles, or generalize to universals. Not all explanatory research is hypothesis-driven, but a significant portion of it is. Exploratory research, by contrast, seeks to discover patterns in observation or data and abstains from drawing strong theoretical conclusions on this basis. It is never hypothesis-driven. Often the aim in exploratory research is building a conception of the phenomenon of interest, defining the research problem, or exploring the hypothesis space.

What matters for our purposes about this characterization of parallel scientific strategies is that no research practices falling under the exploratory banner will involve both something that looks like a canonical theoretical model and something that looks like a canonical data model. These practices frequently involve no modeling at all; perhaps there is only qualitative interviews with patients, an epidemiological case report of the first documented instance of infection by a novel pathogen, or an exhaustive list of behaviors observed in some remote troupe of chimpanzees when first observed by primatologists. Other times, there is only an object that an adherent to the Suppesian account would identify as a data model. The case study in a critique of an exploratory research strategy in single-cell transcriptomics covered in extreme detail in the previous chapter serves to illustrate well the use of models in exploratory research practices. The use of mathematical and computational modeling techniques in the practices critiqued by both Chari and Pachter (2021) and Goddard et al. (2018) are paradigmatically in alignment with philosophical characterizations of data modeling: they are wielded with the objective of eking out patterns in raw data and preparing them for further investigation and analysis, and scientists take themselves to be abjuring from the involvement of theoretical resources in such practices. However, as we witnessed in the exploration of these case studies in the previous chapter, if such modeling practices are to be informative of phenomena in nature, they must involve theoretical resources. As Harris (2003) and Leonelli and Tempini (2020) have argued, all practices of data modeling, and data itself, are theory-laden. Data models, like theoretical models, encode theoretical principles, and must be interpreted using theoretical resources in order to be usable to human scientists. Acknowledgment of this fact already seems to blur any boundary we might attempt to draw between models of theory and models of data along lines of human interpretation and use.

While it is undoubtedly the case that some of the conceptual objections wielded by scientists conform roughly to the archetypes of models of theory and models of data, it is by no means the case that scientific inference necessitates a theoretical model being compared with a data model to proceed. This is, of course, the role that Suppes envisioned for these modeling types in scientific practice. On the semantic view of theories, data models are a necessary component of a research protocol in order for any theory or hypothesis to receive empirical confirmation. As we saw in the discussions of qualities of data and views on representation, however, this depiction of scientific practice failed to account for empirical substantiation: no satisfactory story was ever supplied concerning how data models or data represent or are grounded in reality. And, indeed, when we look to scientific practice, we instead find a plethora of modes of modeling and modes of empirical confirmation. The neat and homogeneous story supplied under the semantic view both failed in its aim to deliver a depiction of empirical confirmation and failed to accord with the observed diversity of scientific practice. There remains, however, one final avenue left to explore: can the distinction between data models and theoretical models play a useful role in indexing the degree of confirmation of a model's empirical content?

3.4 The Confirmation of Empirical Content

Science distinguishes itself in being an empirical mode of enquiry. For a model to count as a scientific model, therefore, it has to have "stuck its neck out;" it must possess empirical content—whether theoretical model, data model, or any other species of scientific model. This means that a model must have content that is either experimentally or observationally substantiated, or it must have content that the world could, in principle, push back on. But models differ considerably with respect to the epistemic status we accord their empirical content. Naturally, if we conjecture some relation or principle which might hold in the world and we codify this in a formal model, this has different epistemic standing from a model encoding values that have been experimentally validated. It is clearly important that we philosophers of science have the conceptual resources to distinguish between the variable empirical content of models and its variable degrees of empirical conformation. The lack of such conceptual machinery makes an accurate appraisal of scientific practices and how they generate knowledge impossible. Further, if philosophy of science has any ambition to provide conceptual resources to working scientists, the ability to draw such a distinction is necessary lest our input be misguided.

The account of models of data as distinct from theoretical models appears, to a first approximation, to perform this function. Theoretical models express content that is conjectural in nature for the purpose of comparison with empirical lines of evidence. Data models, on the other hand, are understood as minimal manipulations of the direct results of observation or measurement procedures which serve to confirm or refute the gambits of a theoretical model. Undoubtedly, were there in reality a sharp divide between models that contained strictly and exclusively conjectural, theoretical content and models that contained strictly and exclusively empirical content, the Suppesian typology of models would be capable of tracking the extent of confirmation of a model's content. No such strict separation holds, however, in the regular work of science. To see that the distinction between models of data and models of theory cannot reliably track differences in the degree of confirmation accorded to empirical content, we need only observe that the models routinely deployed by working scientists contain both theoretically-motivated empirical content that is conjectural in status and content that is more or less directly imported from experiment or observation.

Physics was once a procedure of laborious longitudinal observations carried out by hand or measurement from rudimentary experiment in conjunction with mathematically-aided speculation about the machinations of our universe. Increasingly, the cutting edge of various branches of the physical sciences involves the use of measurement devices whose very process of recording data is model-guided. These models determine where within the field of a telescope to zoom, enhance, or take finer-grained recordings (Rehemtulla et al., 2023). These models also determine which select few, out of readouts from millions of particle collision events, to retain for further evaluation (Belis et al., 2024a). These models work in detection and parameter estimation for gravitational waves in gravitational wave astrophysics for readouts from Laser Interferometer Gravitational-Wave Observatory (LIGO) detectors (George and Huerta, 2018). Importantly, each of these measurement devices itself encodes a model which integrates both theory-driven, conjectural content and data readouts in real time. These examples do not involve the comparison of a model representing theoretical constructs with a model exhibiting the data. In each example, *what data is in fact measured and analyzed* is a product of instantaneous comparison between theoretical principle and live instrument readout. Similar hybrid contents can be found in many computer simulations and in many applications of machine learning in science, such as the AlphaFold 2.0 example examined in detail in the previous chapter.

Any attempt to interpret such hybrid models—which abound in science—in line with standard depictions of either theoretical models or data models forces us to misconstrue the content, epistemic power, and appropriate usage of such models. If such models are presumed to be theoretical models and assessed accordingly, then artifacts of immediate measurement procedures are interpreted as theoretical principles. If, on the other hand, such hybrid models are treated as data models, then conjectured principles are reified as experimental evidence. Indeed, given 1. the empirical origin of our theoretical knowledge, and 2. the conclusions reached in the previous chapter concerning the degree to which data and measurement practices are essentially theory-laden, or structured by human concepts, we seem to be forced to conclude that the content of any scientific model is necessarily both empirically-informed and theory-laden.

The problem of false dichotomization causes trouble even in the use of models that look like standard-issue data models or theoretical models. What we have traditionally classified as data models in actual scientific practice can express much fine grained differentiation in the degree to which their content has been confirmed; the same holds true of models of theory. Moreover, as I hope to have motivated in both the previous section and the previous chapter, all data and data modeling activity is theory-laden. Prior conceptual content, some of it conjectural in nature, makes its way into any modeling procedure, no matter how low to the experimental ground or empirically-driven. The idea that some models are derived with only input from data further serves to reinforce the idea that some scientific procedures can be made "free from theory," which, I hope to have expressed convincingly in the last chapter, is both untrue and harmful to the integrity of scientific best practice.

The Suppesian distinction is purported to do the work of separating models which differ in the nature of their empirical content. Indeed, we might conjecture that the reason this distinction has persisted so long in the philosophy of science literature—while advances in the state of discourse progressively removed each leg it had to stand on—is that it appears to loosely track a distinction in degrees of confirmation of empirical content. I have endeavored to show in this section that the distinction between models of data and models of theory does not sufficiently track the status of empirical content. In papering over this more nuanced property of models, the theory model/data model distinction obscures what is both philosophically important and practice relevant.

Philosophy of science will needs tools within their conceptual arsenal capable of tracking both the manner and extent to which a the contents of a model have received empirical support. A viable alternative to the distinction between data and theoretical models ought to look to comparisons within the contents of a given model, rather than taxonomizing models on the basis of superficial features. For instance, we may believe that we have demonstrated the in-principle possibility of some mechanism by means of simulation or formal result. This hypothetical mechanism may be worked into a model that contains observational data from a limited number of instances. A third model may establish the experimental infrastructure to test for the generality of such a hypothesized mechanism. The objective, in any case, must be to index empirical content by the extent to which it has received confirmation and the manner in which the confirmation has been established. This taxonomy of models and model contents is, notably, much less neat than the framework it would seek to usurp. This, I allege, is an acceptable price to pay for verisimilitude.

3.5 Conclusion

Much as the latter half of the 20th century saw philosophers make the shift from a syntactic conception of scientific theory to a semantic view, which placed emphasis on the notion of a scientific model, recent decades have seen a shift towards what has been dubbed a pragmatic turn in the philosophical conception of theories and models. Advocates of this pragmatic turn have worked to make their revised conceptions of the epistemic and representational function of models compatible with one of the essential distinctions of the semantic view: the distinction between models of theory and models of data. Philosophers of science operating within this pragmatic paradigm speak of data models as distinct from theoretical models, tacitly affirming the compatibility of the essential features of this distinction with their commitments to the nature of modeling and representation.

I have attempted to show in this chapter that the revisions to received conceptions of the ontological, logical, epistemic, and representational faculties of models entailed under the pragmatic view are incompatible with any robust distinction between data and theoretical models, at least in the spirit in which this distinction was originally intended: as a fundamental division in logical type, representational status, and epistemic status. The distinction, moreover, fails to robustly track differences in the practice of using and interpreting models. The important conceptual distinction that must be drawn, I have argued, concerns the degree and extent to which a model's contents have been empirically substantiated. While the Suppesian distinction appears, to a first approximation, to track this property of a model's empirical contents, I argue that it fails to do so systematically. As a result, it can only confuse and obstruct distinctions of real epistemic significance.

Disabusing philosophy of science of the false dichotomy between models of theory and models of data has far reaching implications for classical debates in philosophy of science, for philosophers' engagement with scientific practice, and for the work of science itself. If, to take merely one example, philosophers of science were to put forward a taxonomy along the lines of differences in empirical content, the debate concerning the epistemic status of computer simulations and its comparison to both traditional modeling techniques and experimental regimes is rendered obsolete, as such differences (or lack thereof) are simply subsumed under a taxonomy of empirical content. Allowing for the diversity of empirical content that models employ in actual scientific practice allows for more truthful and nuanced philosophical engagement with scientific practice. Philosophers will be better served refocusing their analytic resources to the parsing of epistemic content in place of classifying models according to their position in a theorized hierarchy which fails to veridically capture scientific practice. Insomuch as scientists heed philosophical characterizations of their work, the negation of a sharp distinction between data modeling and theoretical modeling, alongside clarifications in the status of empirical content, will help researchers avoid fallacious inferences.

CHAPTER 4
SCIENTIFIC REPRESENTATION & CONCEPTUAL REIFICATION

4.1 Introduction

In the recent history of the field, the nature of scientific representation has been one of the most generative—and most hotly contested—topics in philosophy of science. Perhaps surprisingly, philosophers have paid little attention to how scientific representation goes wrong in practice. I refer to the patterns of mistakes that emerge when the interpretation of models misses its mark as instances of *conceptual reification*. Reification occurs when, in the course of interpreting a scientific model, some artifact of the representation is mis-mapped onto a target system, epistemically compromising the empirical inference at hand. I argue that specific features of a philosophical account of representation—namely, whether or not it conceives of representation as mind-dependent—impact our ability to handle reification. "Mapping accounts" of representation, on which representation is understood as a two-place, objective relation, obscure sources of reification. Agent-relative depictions of representation, meanwhile, facilitate our ability to identify, analyze, and correct for reification. Insomuch as commitments to the nature of representation guide scientific use of models, triadic accounts help modelers to avoid reification.

Reification, I submit, is one of the chief epistemic confounds of mathematical modeling in science. Philosophers speak of the difference between the *de dicto* and *de re*; the act of *re*-ification is to mistakenly treat the *de dicto* as the *de re*. It occurs when some accidental facet of the descriptive medium or representational vehicle—or how it is interpreted—are imputed to the phenomena under study. In brief, reification is treating some feature of a mathematical representation that is there for reasons of convention or convenience as real and worldly, in a manner that leads us epistemically astray. Correctly interpreting the results of a mathematical modeling procedure requires the—sometimes painstaking—disambiguation between those features of a model which are intended in a representational spirit or which have been put to the empirical test and those features not of direct epistemic significance.

Reification is, essentially, an error of interpreting scientific representation. Representation is a cornerstone topic within modern philosophy of science, and philosophers have put forward a plurality of accounts detailing how models represent target phenomena in nature to epistemic success. Epistemic failure cases, by contrast, have received little attention, and philosophical characterizations of representation have not historically been measured according to their ability to make sense of the errors that arise in the course of inference with, and interpretation of scientific representations. I suggest here that this is an essential desideratum for a viable account of scientific representation. My rationale for doing so is first, the recognition that failure cases in science are often as informative as success cases, and a characterization of science on the basis of its successes alone often generates a misleading picture. In the second place, only an account which can accommodate failure cases is able to make sense of scientific practice in such a way as to facilitate corrective measures.

I will argue here that one axis carving up the space of philosophical positions on representation, in particular, impacts our ability to handle errors of reification as they arise in scientific practice. This is the axis which runs from *mind-independent* or to *mind-dependent*. Mind-independent accounts of representation are *dyadic*: they hold representation to be a relationship which holds objectively between objects or structures. Mind-dependent or agent-involving accounts, meanwhile, are *triadic*: the interpretations or intentions of an epistemic agent is taken to be an essential element of what it is to represent.

I will defend in this chapter the thesis that dyadic accounts of scientific representation pose an obstacle to our ability to identify conceptual reification. Accounts of representation which view representation as a three-place relation—essentially evolving a human epistemic agent—conversely make it easier to address reification. Along the way, I seek to offer a more concrete account of conceptual reification than has been offered previously in the (limited) philosophical treatment of the subject. I also put forward a taxonomy of ways in which reification comes about in different modeling exercises. Through a series of case studies, I illustrate how our stance on representation—specifically, whether we take it to be mind-dependent or mind-independent—changes how we evaluate instances of reification in scientific practice.

4.2 Accounts of Representation

Philosophers of science seek to characterize how the tools and methods of science enable us to learn about the world. The essential conceptual instruments of science are theories and models. The epistemic purchase of these tools over nature has been predominantly understood in representational terms by the philosophical literatures that have addressed them (de Oliveira, 2021b). For much of the history of these literatures, the representation relation between model and world was taken to be mind-independent (Frigg and Nguyen, 2017b; Knuuttila, 2010; de Oliveira, 2021b). In recent decades, the practice turn in philosophy of science has seen philosophers more attentively looking to the quotidian details of research practices and conceptualizing of science as a fundamentally human activity (Frigg and Nguyen, 2017b; Knuuttila, 2010; de Oliveira, 2021b). The emerging picture admits of the diversity, messiness, and opportunism of modeling strategies and, above all, the essential input of human reasoning.

The shift from one attractor of consensus to the other was subtle and distributed, and the consequences of this newly-acknowledged observer-relativity have yet to be fully explored. In particular, the exigencies of an account involving an ineliminably *cognitive* role have yet to be worked out for an account of the operation of models in science, in particular, I urge, for the errors that arise in modeling work.

de Oliveira (2021b) and Frigg and Nguyen (2017a) make note of this consequential shift even within the works of particular scholars in the field. Giere (1988), with his similarity account, and Van Fraassen (1980b), with his isomorphism account, were among the original proponents of canonically mind-independent accounts of representation. By the early 2000s, Giere had much revised his account. While still endorsing a version of similarity, he came to place emphasis on the agent-relativity of scientific representation (Giere, 2004, 2010). Van Fraassen similarly has come to view the representational status of models as a matter of use by an agent towards a particular epistemic end (Van Fraassen, 2008). A slew of more recent accounts of modeling have started out on observer-relative footing, including flavors of inferentialism (Suárez, 2003; Contessa, 2007), fictionalism (Frigg, 2010; Frigg et al., 2020), and novel accounts of similarity (Weisberg, 2007, 2013; Godfrey-Smith, 2006), although the last of these has along its own trajectory brought the agent-relativity of representation to the fore as the account developed. Artifactualism (Morrison, 1999; Morgan and Morrison, 1999; Knuuttila, 2011) and antirepresentationalism (de Oliveira, 2021b; Grüne-Yanoff, 2013; Batterman, 2010) and pluralism (Downes, 2011) take a user-centric approach to understanding the epistemic role of modeling in science which de-emphasizes or else outright denies the representational role.

Most philosophers of science now endorse mind-dependent accounts of representation, including several of the foremost proponents of dyadic accounts, now reformed. This subtle shift in consensus to mind-dependent views on representation was partially spurred by a more general shift in philosophy of science towards an acknowledgment of science as a fundamentally human activity. It has also been motivated on the grounds that dyadic accounts failed to meet core desiderata for an account of representation: satisfying the logical properties of representation (Bailer-Jones, 2009; Goodman, 1976; de Oliveira, 2021b; Frigg and Nguyen, 2017a; Suárez, 1999).

Goodman (1976) delivers a philosophical analysis of representation and its logical properties, arguing these to be unsatisfied by any account that seeks to ground representation in resemblance or similarity. Goodman characterizes a "naive view" of representation as any position that holds that "'A represents B to the extent that A resembles B.'" "Vestiges of this view," Goodman writes, "with assorted refinements, persist in most writing on representation" (Goodman, 1976, p.3). As Goodman argues, the resemblance relation is reflexive, symmetric, and transitive. Representation, meanwhile, is none of these: "An object resembles itself to the maximum degree but rarely represents itself; resemblance, unlike representation, is reflexive. Again, unlike representation, resemblance is symmetric...Plainly, resemblance in any degree is no sufficient condition for representation" (Goodman, 1976, p.4)

Suárez (1999) adapts and extends Goodman's arguments to the puzzle of *scientific* representation, building a case against the two most popular views on representation within the family of dyadic or mapping accounts: similarity and isomorphism. Suárez recapitulates Goodman's assessment of the logical properties of representation, judging similarity and isomorphism to contradict these requirements: "[r]epresentation in general is an essentially non-symmetric phenomenon...[r]epresentation is also non-transitive and non-reflexive (as a matter of fact representation is probably in addition irreflexive, asymmetric and intransitive)" (Suárez, 1999, 232).

Suárez further argues that additional features beyond similarity or isomorphism are necessary in order for something to count as a representation or something else. For any two objects, no matter how facially unrelated, we could identify an open-ended number of respects in which these are similar or share in logical properties, (e.g., we can predicate physical existence of both, both exist within our known universe, etc.). Perhaps the most damning line of attack against mapping accounts is there inability to handle *mis*representation.

The possibility of misrepresentation is widely taken to be definitional of representation relations, hence a desideratum for any adequate philosophical account thereof (Dretske, 1986; Fodor, 1984; Van Fraassen, 2008; Millikan, 1990; Park, 2014; Shech, 2015; Suárez, 2003, 2004). This is the case whether the representation relation is one of artistic representation, mental representation, or scientific representation; hence the literatures in aesthetics, philosophy of cognitive science, philosophy of mind, and philosophy of science have all rallied around fundamentally shared desiderata for a theory of representation Park (2014). It is not merely that a philosophical account of representation has to accommodate the *evident fact* of misrepresentation in one or another domain, but rather that the possibility of misrepresentation is held to be a constitutive requirement of representation. If the relation picked out by a philosophical account does not accommodate the capacity for misrepresentation, then the relation picked out is, definitionally, not a representation relation; hence the philosophical account proves to be an account of something other than representation.

"A capacity to misrepresent," write Schulte and Neander (2012), "is often thought to be essential for representing: no possibility of misrepresentation, no representing" (p.1). L'Hôte (2010) writes: "the phenomenon of representation carries the possibility of misrepresentation along with it" (L'Hôte, 2010, p.1). "Any adequate account of representation must be able to explain the phenomenon of *misrepresentation*. This follows from the normative character of representation." (Rowlands, 2011, p.119). "A satisfactory account of what constitutes the relation of representation," Sánchez-Dorado (2025) argues, "ought to be able to accommodate the persistent phenomenon of misrepresentation" (Sánchez-Dorado, 2025, p.62). Frigg and Nguyen (2021) attest that "allowing for the possibility of misrepresentation is a key desiderata required of any account of scientific representation."

The ability to account for misrepresentation is deemed essential for the project of characterizing scientific representation. Suárez divides misrepresentation into two substrata: *mistargeting* and *inaccuracy*, arguing that mapping accounts can accommodate neither. The latter of these is important for an account of scientific representation to handle the ubiquitous phenomenon of *idealization* within scientific models. The former of these speaks directly to our thesis. Suárez (1999) defines mistargeting as "mistaking the target of a representation" (Suárez, 1999, 233). Importantly, the capacity of a theory of scientific representation to handle the fact of *misrepresentation*, in particular, *mistargeting*, is a necessary precondition for it to account for *reification*. We have defined reification of a scientific model, recall, as the *mismapping* of incidental facet of the descriptive medium or representational vehicle—or how it is interpreted—onto target phenomena in nature. Suárez offers an example of mistargeting by way of mistaking the identity of the subject of a painting: "This is a clear case of misrepresentation," Suárez urges, "but there is no failure of similarity to explain it. Indeed misrepresentation by accidental similarity would be impossible if [mapping accounts] were true, precisely because similarity would then warrant representation. Exactly the same argument goes *mutatis mutandis* for isomorphism, and it is an argument that can be easily transferred to cases of scientific mistargeting" (Suárez, 1999, 234).

The extension of the concept to the scientific domain speaks immediately our concerns with the ability to portray reification: "a mathematician's discovery of a certain new mathematical structure (defined by a new equation perhaps) isomorphic to a particular phenomenon would amount to the discovery of a representation of the phenomenon—independently of whether the mathematical structure is ever actually applied by anyone to the phenomenon. The history textbooks would have to be rewritten so that it was Riemann, not Einstein, who should get credit for first providing a mathematical representation of space-time" (Suárez, 1999, 234). This is because, on mapping or dyadic views on representation (such similarity and isomorphism), the representation relation is *intrinsic* to the relation between representandum and representans; it is *out there*, *given*, *objective*, and available to *discovery*.

If relationships between models and the worldly systems they are taken to represent are understood as objective and mind-independent, the very concept of reification cannot be made intelligible. An epistemic agent cannot err by proceeding from a mistaken mapping between mathematical structure and worldly structure, for that mapping is not stipulated, but discovered as a brute fact. A rigorous grounding of the constructs of a model is likewise superfluous if models inherently represents physical structure in the world. According to dyadic accounts of scientific representation, models represent their targets in virtue of objectively-available similarities or shared structure. Dyadic views on representation, I argue, makes it difficult to furnish an account of how a user or interpreter of a representation could arrive at a mistaken mapping between a representation and a target. The very idea that an epistemic agent could establish a *mistaken* mapping between representational vehicle and target implies that epistemic agents are capable of establishing mappings between representational vehicles and targets and, moreover, that the choices, activities, and interpretations of said agent are what is epistemically salient. Once an agent and their intentions or the sense they make of a model is essential to the epistemic work of wielding a model (representationally), we have shifted from the purview of a dyadic account to a triadic or "hybrid" account, as we are positing an essential epistemic role for a human agent in the business of representation.

Critically, dyadic accounts further fail to make sense of misrepresentation in classifying all mistaken mappings or non-veridical representations as nonrepresentations. Mapping accounts, such as isomorphism, fail to meet an essential "condition of adequacy" for a philosophical account of representation, namely "that misrepresentation must be possible" (Frigg, 2006, p.59). Frigg continues, "To require that a model must be isomorphic to its target amounts to saying that only accurate representations count as representations and to ruling out cases of misrepresentation (i.e. cases in which isomorphism fails) as non-representational, which is unacceptable" (Frigg, 2006, p.59). Mapping accounts have "difficulty distinguishing between misrepresentation and non-representation" (Frigg and Nguyen, 2021). "Asking what makes a representation an accurate representation," Frigg continues, "*ipso facto* presupposes that inaccurate representations are representations too. And this is how it should be. If S does not accurately represent T, then it is a misrepresentation but not a non-representation. It is therefore a general constraint on a theory of scientific representation that it has to make misrepresentation possible" (Frigg and Nguyen, 2021).

The literature concerning models and their representational function in philosophy of science now broadly recognizes the inability to accommodate misrepresentation as a nail in the coffin of mapping accounts. Moreover, just as in accounts of mental representation, it is made clear that this is a shortcoming particular to *dyadic* views on representation: "Misrepresentation makes it clear that representing is often a three-place relation. Suppose, for example, that I see some crumpled newspaper blown by the wind as a cat slinking down the street. There are at least three things involved. First, there is the representation (or representational vehicle) that has the content...Second, there is the thing that the representation is aimed at representing, in this case this is the newspaper. Cummins (1996) calls this the target of the representation. And third, there is the content of the representation. Since I represent the newspaper as a cat, the content of the representation in this case is cat. Misrepresentation has occurred in this case because the target of the representation is not in the extension of the representation; the newspaper is not a cat" (Schulte and Neander, 2012). Were representation a mind-independent, two-place relation holding between representational vehicle and referent, a mistaken referent would be impossible. Misrepresentation could not obtain.

Conceptual reification is an inferential fallacy particular to the interpretation of scientific models. It is an error that comes about by misrepresentation. On dyadic, or mapping accounts of scientific representation, misrepresentation cannot be made sense of as representation at all. How, then, does reification come about, what does it look like in scientific practice, and how do our commitments to the nature of representation play out in the identification and handling of reification? I address these matters in the following section.

4.3 Reification

Alfred North Whitehead called it "the fallacy of misplaced concreteness." It has been referred to variously as "conceptual reification" (Millstein & Skipper, 2005), "pernicious reification" (Winther, 2019), "hypostatization" (Sober, 1980), and simply "reification" (Levins & Lewontin, 1985). The phenomenon of mistaking accidental features of formal models for features of the systems they are wielded to model—to, at times, catastrophic effect—has been noted everywhere from quantum mechanics to the quantitative social sciences.

To reify is to treat the merely hypothesized, stipulated, or abstract as real, worldly, and concrete. Reification of a formal model occurs whenever we treat merely incidental features of a formal model as features of the systems they are wielded to model. Sometimes this happens quite straightforwardly, when we take features of a theoretical model which have mere heuristic value as true of nature, or when we treat them as having undergone empirical validation in wielding the model when only unrelated features of the model have been put to the test. It can happen more subtly, however, in the failure to identify how much assumption, how much prior conceptual commitment to the nature of the system under study, goes into the most simplistic of formal analyses. Importantly, for reification to be pernicious, for it to lead us astray epistemically, this conceptual cross-pollination must ultimately result in a misapprehension of the nature of the phenomenon under investigation.

Reification, broadly construed, is an umbrella-concept for the kinds of conceptually-driven epistemic errors scientists are liable to make in wielding mathematical methods. Of course, many things can go awry in applying mathematics to the world. Most of these difficulties, however, are problems that scientists and mathematicians are perfectly capable of addressing on their own—they are trained to handle errors in calculation or the miscalibration of instruments. Uniquely conceptual errors, however, those that lead to reification if not discovered and addressed, often require outside intervention. Propitious, then, that a field of scholars is trained in the laborious conceptual disambiguation of scientific activities.

Reification, it would seem, is perhaps the most obvious place for philosophers of science to intervene on modeling practice. Thus one of, if not the, primary foci of an applied philosophy of mathematics ought to be reification.

Interestingly, the pragmatist philosopher C. S. Peirce believed something like reification to be an indispensable component to the effective use of mathematics in learning about the world, although he used the term *hypostatisation*. "Mathematical reasoning," Peirce contended "almost entirely turns on the consideration of abstractions as if they were objects" (Peirce, 1958, 509). Although we may regard a stipulative treatment of mathematical structure as real or concrete as an often useful step in the application of mathematics to nature, we must ultimately, at the final stage of analysis recognize that we have engaged in this step of reification or hypostatisation and perform conceptual disambiguation in order to know what our modeling efforts in fact permit us to know.

There are two broad directions in which reification occurs. These, in turn, correspond to two extremes of misinterpreting the representational nature of mathematical modeling methods. The first direction of reification I dub *intuition laundering*. This is supported by adherence to a policy of naïve empiricism, in which the modeler takes themselves, to paraphrase from Lewontin, to be allowing nature to speak to them through data in a theory-free manner. Theoretical analysis performed at the outset of the modeling procedure is superstitiously believed to taint the process; data-modelers pursue a mythos of empiricism unburdened from the biases of prior knowledge or human conceptual grasp. I have called the corresponding ideology—that we ought, as much as possible, to abjure from theorizing to avoid tainting our empirical efforts with preconception—a *theory-free ideal*, after Douglas' *value-free ideal*, as introduced in Chapter 3. This attitude, of course, engenders reification: an inferencing procedure cannot get off the ground without the importation of conceptual resources; without some, however primitive, background ontology or conceptualization of that which falls under our investigation. Attempts to do science unbridled from the conceptual commitment of theory simply blinds us to the nature and extent to which our intuitive prior conceptualizations and theoretical commitments shape the outcome of our empirical work. The modeling effort then functions as a kind of Rorschach test: we are free to read into it whatever we please while congratulating ourselves for our objectivity. The use of analysis of variance, principle component analysis, and linear and multiple regression, as critiqued by Lewontin (1974) (1974) exemplify this first genre of reification.

Intuition laundering and the theory-free ideal occur in data-heavy sciences with the use of what philosophers of science have dubbed "models of data," in contrast to theoretical models or models of theory (Suppes, 1962a; Jacquart, 2016). Data modeling practices are understood as mere curve-fitting exercises. It is assumed that the use of mathematical methods in such applications is quite radically different from the more theoretical modeling, in that data models do not represent features of their target systems. As we have covered in Chapter 3, this dichotomization of modeling practices in this way appears to be both false and harmful. The application of mathematics to the study of nature of any form and to any epistemic end requires mathematical representation and some degree of commitment to the nature of the target phenomenon.

A second misconstrual of modeling practice enables reification in granting theoretical significance or realism to the wrong elements of a model. Whereas in so-called "data modeling" exercises it is presumed that nothing of the mathematical system employed represents features of our target phenomenon, in theoretical modeling it is sometimes presumed that every aspect of the mathematical system at hand must act as a direct and truth-apt abstract representation of features of the phenomenon. But there are always myriad aspects of a target phenomenon not captured in a formal representation. The very conceptualization of a phenomenon of interest itself distills cognitively-salient features of our perceptible universal from the vast, entangled milieu of physical reality. Mathematical systems, further, always contain both idealized and strictly conventional elements. The mapping between math and territory, as it were, is by no means straightforward. In taking every component of a theoretical model to be a direct and literal representation for a target phenomenon, we reify elements of our formal representation, hence leading our scientific efforts astray.

Finally, it is common to see reification come about in the process of model transfer. In an initial modeling context, a model is built and rigorously grounded in the conceptual apparatus of a supporting theory and in what is empirically known or discoverable according to the measurement protocols of a specific scientific regime. Together, this grounding should license sound scientific inference. Model transfer involves taking the formal infrastructure of a model from its initial domain of application and applying it to new phenomena. This typically leaves the core formal architecture of the model, the representational vehicle, largely unaltered, but switches the intended referent of the constructs therein. Model transfer is a perfectly legitimate—and often fruitful—strategy in the sciences, when approached with care (see., e.g., Potochnik (2012)'s discussion of model transfer from economics to evolutionary game theory). It is not uncommon, however, for the interpretation of a model within the new domain to display conceptual baggage from its initial domain of application. This vestigial interpretive structure has not received conceptual or empirical support in this new domain of application; if it happens to hold true of the phenomena under study, it is only luckily so.

4.4 Case Studies

In this section, I run through a series of examples with the intent of showcasing the phenomenon of reification in scientific modeling practice and its heterogeneous etiology. It is my hope that an exploration of these case studies will serve two essential functions. In the first place, I hope in these case studies to exhibit the different forms of reification. In the second place, I use these exemplars as the grounds on which to illustrate how differing stances on representation shift our perspective on the modeling exercise and can make the error variously impossible to discern and make sense of, as in the dyadic accounts, or readily apparent.

4.4.1 Classical Reification of Theoretical Constructs

Bangu (2008) describes how an essential breakthrough in quantum mechanics, the discovery of the $\Omega-$ baryon, was formally and theoretically motivated, despite looking like an erroneous inference from a problematic reification of formalism. After the revolutionary breakthroughs in early-20th-century physics, the mid-century brought with it a steady stream of newly discovered subatomic particles, with theoretical work continuously pointing to yet more undiscovered species. Known as the "particle zoo" era, which was considered a chaotic time in physics, lacking in strong theoretical guiding principles that would unify the search for new fundamental particles.

Such a unified approach was simultaneously introduced by both Gell-Mann and Ne'eman, in what Gell-Man was to call the "eightfold way": a classification scheme based on the observation of symmetries. This scheme not only introduced order to the chaos of the particle zoo, it led to the prediction of novel particles like $\Omega-$, and was a a necessary precursor to the development of the quark model.

The eightfold way sought to organize particles first relative to their status as baryons and mesons, then by spin. When these particles are then plotted by charge and strangeness, a symmetrical overarching pattern appears. Gell-Mann and Ne'eman's inference proceeded from the observation that nine places of the decuplet drawn up representing baryons with spin $-\frac{3}{2}$ were already known particles. The tenth place was an unknown, a blank. Of course, were such a particle to exist, it would have to be a baryon with charge –1, strangeness –3 and a mass near 1.67 GeV, or 1680 MeV/c2. Gell-Man inferred that the empty space indeed should be occupied by a real, as yet undiscovered baryon, christening it $\Omega-$. In 1964, a particle with precisely those characteristics was deduced from cloud chamber traces in the laboratory at Brookhaven.

The puzzle Bangu (2008) presents us with is this: what licensed treating that fill-in-the-blank chart as necessitating the existence of a real subatomic particle with such properties? As Bangu understands it, the inference "requires an additional premise—which I'll call the 'Reification Principle'." Bangu defines this principle of reification as follows: "If Γ and $\Gamma\prime$ are elements of the mathematical formalism describing a physical context, and $\Gamma\prime$ is formally similar to Γ, then, if Γ has a physical referent, $\Gamma\prime$ has a physical referent as well" (Bangu, 2008, p.248).

Bangu does not specifically characterize the issue of interpreting this episode in the history of physics in terms of antecedent commitments to the nature of scientific representation. Importantly, however, Bangu is only able to identify the puzzle presented in this instance—the puzzle of reification—in light of an understanding of formalism as representing physical structure in the world *only in virtue of a human interpreter*:

> "[T]he crucial reificatory step (the indication of the new particle's characteristics) was taken by interpreting the mathematical formalism. Hence, it is not the formalism itself that predicts, as it were, but our interpretation of it: as is usually the case, without a physical interpretation, no empirical predictions could ever be obtained from any formalism. This observation is meant to cast doubts on the claim that something really novel took place in the omega minus episode" (Bangu, 2008, p.254-255).

The propensity to reify formalism, imputing physical structure to formal elements in the absence of any substantive theoretical or empirical rationale, is "at odds with the standard naturalistic, empirically oriented, scientific methodology," according to Bangu; "there is no naturalistic explanation why [the reification principle] could work, while the very idea of a higher-level correspondence between mathematical objects and reality reminds us of numerology, astrology, and other dubious practices involving reifying mathematics" (Bangu, 2008, p.250).

If the inference-licensing representation relation between mathematical structure and the physical reality consisted in a brute sharing of structure or resemblance, the "correspondence between mathematical objects and reality" Bangu identifies would be real, discoverable, and sufficient to license inference over that physical reality. On the agent-relative conception of mathematical representation held by Bangu, relations between formalism and reality are meticulously established by scientists. Inference to reality based on observation of structural similarity within a formal system alone is, therefore, reification.

The inference to the existence of the $\Omega-$ baryon was undoubtedly lucky. It was not, however, an altogether fallacious inference. Recognizing this fact requires recognizing that the inference was grounded on more than mere "formal similarity." In the context of the eightfold way schematic, formal similarity and positioning within the schema were themselves the produce of a deeper theory—albeit a theory which remained, at the time, partially enshrouded in mystery by even its progenitors. In other words, the inference to the physical reality of $\Omega-$ baryon was not a matter of formal similarity alone, but a matter of the conceptual role played by those similar formal objects within an overarching theoretical framework.

4.4.2 Extending Psychological Predicates

A recent bio-, neuro-, and psychological sciences has seen researchers pursuing the investigation of mentalistic properties in a range of systems beyond those normally considered psychological, such as fruit flies, neural networks, individual neurons, or plants. The object of research is behavior requiring some degree of internal processing and awareness of the environment, for instance, memory and learning. Extending the reach of paradigmatically human cognitive faculties to systems that are not normally thought of as cognitive is a promising horizon of research, and one with important real world consequences for how we understand and interact with such systems.

Figdor (2018) looks to combat what she views as a problematic anthropocentrism in our attribution of psychological predicates. She views the applicability of formal models originally developed for applications to cognitive processes as grounds for extending these concepts to systems that might not intuitively strike us as cognitive. "[A] mathematical model," she writes, "provides strong evidence that two domains have important similarities" (Figdor, 2018, p. 135).

The temporal difference model, introduced into the conditioning literature in 1987 by Sutton and Barto (1987), is a mathematical model of conditioning, usually observed in a cognitive organism. More specifically, the temporal difference model represents the association strength between a conditioned and unconditioned response over time. A paper by Suri and Schultz (2001) presents the results of training a neural network with the temporal difference model. The resultant model is proclaimed to replicate firing patterns observed from electrophysiological recordings in monkeys, and taken as evidence of a shared capacity for learning.

Figdor (2018) summarizes Suri and Schultz (2001)'s findings as follows: "use of the [temporal difference] model in a network simulation showed that real neuronal populations appear to be adaptive elements that learn to predict future rewards in the quantitatively similar sense that humans, monkeys, rats, and other adaptive elements do." (Figdor, 2018, 53). Figdor champions the Suri and Schultz (2001) result as "strong empirical evidence that real neurons are adaptive elements in a very rigorous sense" (Figdor, 2018, 54).

In a critical review of Figdor's work, however, Gamboa (2024) problematizes her reliance on *quantitative similarity* to ground the extension of psychological predicates to non-canonically cognitive systems. Elsewhere, Figdor appears to take this notion of quantitative similarity to consist in a sharing or instantiating of formal structure, writing that inference is a matter of "finding structure in neural behavior that is quantitatively analogous to the structure of reinforcement learning in a behaving animal" (Figdor, 2018, 53-54). In a philosophical interrogation of Figdor's premises, Gamboa writes that "[d]escribing quantitative similarity in terms of shared formal structure trades one concept in need of operationalization for another. Without further explication of 'formal structure', it brings us no closer to determining when behaviors from systems in different taxa are quantitatively similar enough to infer shared cognition" (Gamboa, 2024, p.7).

That the mere applicability of some mathematical model to disparate systems constitutes strong empirical evidence of shared mechanisms rests on a number of questionable assumptions, several of them empirical. How a scientist goes about operationalizing the constructs of a model in an empirical exercise, how they select which parts or processes of said system are appropriate targets of comparison and at what level of abstraction or coarse-graining to parse them, and how they undertake the measurement procedures necessary to fit the model with data are all applied, empirical issues; issues which could make the difference between such a comparative exercise being epistemically rigorous or clownishly arbitrary.

Other issues with this process, however, are fundamentally philosophical. Gamboa notes that, according to certain accounts of the ontology of mathematics and its representational properties, "mathematical structure is a feature of models but not their targets" (Gamboa, 2024, p.7). These background philosophical commitments render "[s]o-called discoveries of mathematical structure in behavior...just conceptual reifications" (Gamboa, 2024, p.7). What is essential in this case study for our purposes is that the imputation of mathematical structure to worldly systems only counts as a conceptual reification given two essential background philosophical commitments. Gamboa (2024) notes that one of these is metaphysical: whether mathematical structure has mind-independent existence or can be said to inhere in nature—say, in the firing dynamics of populations of neurons—requires antecedent commitment to a particular ontological characterization of mathematics. The other philosophical commitment concerns the nature of scientific representation. Figdor takes the applicability of a model m to distinct natural systems a and b to be strong evidence of similarity or shared structure between a and b. This inference proceeds transitively, for Figdor takes the essential epistemic relation between the model m and each target system a and b in its own right to consist in similarity or shared structure.

In the philosophy of science, accounts which take the relation between model and target to consist in shared structure or similarity have been collectively dubbed "mapping accounts," and share in treating the representation relation as dyadic and mind-independent (Batterman, 2010; de Oliveira, 2021b; Pincock, 2011). That Figdor is committed, at least implicitly, to such a view is evidenced not only by her appeal to similarity or shared structure between model and target, but also by her conceptions of such structural similarities as "discovered."

The opposing conception of representation holds representation to be agent and goal-relative. On such accounts, models only represent target systems, and license inferences over them, in virtue of a human agent electing to use them for this purpose relative to a particular epistemic goal and a particular research context. On agent- and goal-relative accounts of representation, the constructs which make up a model must receive rigorous conceptual *and* empirical grounding in each context in which they are set to license scientific inference.

4.4.3 Hamilton's Rule

Charles Darwin's view of the natural order is a Malthusian one—devised by analogy to the economic theory—in which self-interested individuals struggle against one another for access to insufficient resources. Against such a backdrop, the emergence and evolutionary stability of altruistic behavior has long puzzled biologists. One prominent explanation evolutionary theorists have proposed is that organisms act to increase the fitness of those with whom they share a high proportion of genetic material: a theory known as kin selection. Kin selection is thus taken as 'explaining away' altruistic behavior as backdoor selfishness. Hamilton's rule is an equation used by evolutionary biologists to convey the dynamics of kin selection (Hamilton, 1964).

Two Expressions of Hamilton's Rule

$$rB > C$$

$$rB - C > 0$$

Where C refers to *fitness costs incurred in a prosocial action*, B refers to *benefits conferred upon a conspecific*, and r is the *coefficient of relatedness between actor and conspecific.*

For any act of apparent altruism, Hamilton's rule expresses the relationship between a metric of genetic relatedness between agent and patient, expected benefits to the patient, and expected costs to the agent. Reproductive costs incurred by the actor must be less than the reproductive benefits conferred upon the recipient of the prosocial act, scaled by a coefficients of relatedness between the two. The essence of the rule was memorably expressed by J. B. S. Haldane: "I would jump into the river to save two brothers or eight cousins".

The usage and significance of Hamilton's rule has been heavily disputed. The debate came to a head in 2010 with a paper by Nowak, Tarnita, and Wilson arguing against the utility of Hamilton's rule (Nowak et al., 2010). Nowak et al forward both claims to the effect that Hamilton's rule rarely, if ever, holds true of nature, and also that it or its derivation represent little more than a tautology, a mathematical truism. Nowak et al's contention is that a formulation of Hamilton's rule which is dubbed the "exact and general" version is taken to predict the change in population average trait value but depends on measures of the same as input—precisely because fitness costs and benefits seem to rely circularly on trait value.[1] Nowak et al. conclude that Hamilton's rule is not an expression of a natural rule but merely a statement about the nature of regression coefficients.

1. The apparent circularity of fitness and fitness-dependent formulations of evolutionary dynamics is a recurrent and thorny problem in the formal Darwinism project.

Philosopher Jonathan Birch (2014) wades into this debate in an attempt to offer some clarity (Birch, 2014). To make sense of Nowak et al's claims, and of the fierce debate over the nature of Hamilton's rule more generally, Birch forwards a conceptual boundary between what he dubs Hamilton's rule general and Hamilton's rule specific. Hamilton's rule general is derived from the Price equation augmented by linear regression. The specific version of the principle is derived, instead, from evolutionary game theory—in particular, the one-shot Prisoner's dilemma with assortativity. In the specific version of Hamilton's rule, the likelihood of being paired against a conspecific who will opt for the same game strategy (defect or cooperate) acts as an analogy for coefficients of relatedness. The general version of Hamilton's rule, unless supplemented by additional population-specific data, must hold true of any population subject to selective pressures.

Hamilton's rule, on its specific interpretation, has been critiqued as grossly misrepresenting the range of potential strategies available to an agent, chief among its conceptual issues. Birch associates Nowak et al's claim that Hamilton's rule "almost never holds" with this version of the principle for representing kin selection. Hamilton's rule general, meanwhile, at least in the absence of additional data from real biological systems, is too generic to be predictive, and hence carries the charge of being tautological. Birch further draws a conceptual distinction between Hamilton's rule general taken as an explanatory principle in evolution and Hamilton's rule as mere mathematical statement. While these are inseparable in practice, the conceptual distinction between vehicle and content enables us to disambiguate various uses, claims, and criticisms made of Hamilton's rule: the extent to which Hamilton's rule is meant to be read as generating causal claims and degree of empirical content accorded to the principle is quite variable.

Without a conceptual partition between representational vehicle and representational content, the corrective Birch offered could not have been made. This separation relies on a conception of formal model architecture as intrinsically *devoid* of any meaningful or inference-licensing relationship to target systems in nature. On a dyadic or mapping view of representation, this intrinsic structural similarity or resemblance relation is posited as the essential epistemic relation involved in modeling. On agent-relative depictions of representation, by contrast, the interpretation of a model by an epistemic agent is what lends it its content and epistemic utility. An agent-relative view of representation, therefore, is necessary in order to draw the very distinction that rescued the debate from unending confusion.

4.4.4 Optimality modeling

Angela Potochnik (2007, 2009, 2010) evaluates the status of optimality modeling at large in the biological sciences. In a 2007 paper, Potochnik engages the allegation that optimality models and game-theoretic models in evolutionary biology are a kind of blunt instrument, an epistemic crutch emblematic of immature sciences to be replaced by more sophisticated forms of enquiry as the science matures (Potochnik, 2007). Lewontin (1979) argues that the methodological maturation of evolutionary biology will render optimality models obsolete (Lewontin, 1979). In response, Potochnik forwards reasons to think optimality models are a permanent fixture of the biological sciences (Potochnik, 2007). Namely, provided the right background conditions hold ensuring their applicability, optimality models may furnish the best explanation for the persistence or nature of some phenotypic trait.

Potochnik (2009) challenges a second prominent account of the role of optimality modeling in evolutionary biology: namely, the inextricability of optimality modeling strategies from the adaptationist program (Potochnik, 2009). In countering this claim, Potochnik distinguishes between weak and strong articulations of the optimality modeling paradigm. Conceptions of the use of optimality modeling differ both in terms of the types of modeling gambits they involve and the strength of the epistemic claims they can support.

A weak conception of the epistemic utility of optimality modeling holds the practice to be revelatory of just the influence of selection on the trait or traits in question (Potochnik, 2009). The only strong desideratum for the validity of this exercise is that it represents the role of natural selection accurately. Compare this to the strong usage, which seems to rely on adaptationist assumptions, namely, 1. that natural selection is the only force or only relevant force shaping phenotypic traits, and 2. that over phylogenetic timescales, natural selection perfects or optimizes phenotypic traits. While the prospects of the adaptationist program and the strong use of optimality modeling appear inseparable, the weak use stands on independent grounds. Potochnik urges that the most prevalent usages of optimality modeling in biological practice are those with tempered ambitions, consistent with her articulation of its weaker usage (Potochnik, 2009).

In a 2010 paper, Potochnik investigates the infrastructure of mutual epistemic support between modeling strategies and lines of confirmation surrounding the use of optimality modeling in evolutionary biology (Potochnik, 2010). Natural phenomena, especially biological phenomena, often exhibit complex causal structure. To gain an epistemic foothold, we must deploy a web of interdependent strategies of investigation. Potochnik argues that optimality modeling strategies are not informative or explanatory in a vacuum but instead are reliant on auxiliary modeling practices which do not, themselves, invoke optimality assumptions (Potochnik, 2010). Yet, countering prominent detractors of optimality modeling, Potochnik maintains that the explanatory utility of optimality modeling does not reduce to the more empirically-substantiated or mechanistically-rich practices which lend such models evidential support.

Disentangling various interpretations of optimality models and the gambits they involve from one another, and from the theoretical frameworks in which they were developed rest essentially on understanding the *idealized nature of models.* Dyadic accounts of representation, as Suárez (1999) has urged, have difficulty accommodating idealizations and modeling gambits, or any inaccuracy in a model whatsoever. A proper understanding of the methodological role of heavily idealized models requires acknowledging the essential human interpretive role in the modeling effort, which can involve the strategic use of simplifying assumptions or counterfactual stipulations. The move from models which are not literally true of nature and, hence, could not map veridically onto reality, to accurate knowledge of the world is a further interpretive step. Agent-relative accounts of representation account for both the utility of idealized models and the essential role of human interpreters in putting such models to use.

4.4.5 ANOVA, Regression, & PCA

Richard Lewontin (1974) famously offered a critique of the use of analysis of variance (ANOVA), principle component analysis (PCA), and linear and multiple regression as used in social science and population genetics(Lewontin, 1974). An ANOVA is typically employed to test the explanatoriness of different groupings of data or causal hypotheses by analysing differences across groups in mean variance. It assumes that datapoints are independently drawn, normally distributed, and homoscedastic (i.e., that the magnitude of the variance, or error term, is constant across values of the independent variable or variables).

Lewontin was concerned about the tautological nature of the use of analyses of variance and related techniques; that the procedure of applying such methods of quantitative analysis is responsible for creating the kinds of partitions and relationships in data it is putatively wielded to discover. All the while such methods are touted as being maximally objective and empirical.

> "The analysis of variance is a tautological partitioning of total variance among observations into main effects and interactions of various orders. But as every professional statistician knows, the partitioning does not separate causes except where there is no interaction...Yet natural and social scientists persist in reifying the main effect and interaction variances that are calculated, converting them into measures of separate causes and static interactions of causes." (Lewontin,

1974, 155)

In carrying out and publicizing their analyses, researchers repeat ad nauseam that correlational methods should not be given a causal gloss without substantiating evidence. In reality, Lewontin urges, we would not be performing and publishing the results of said analyses were we not intending them to be interpreted causally. Lewontin refers to the epistemic errors that practitioners generate in carrying out brute-force statistical analyses as varieties of reification.

> "The most egregious examples of reification are in the use of multiple correlation and regression and of various forms of factor and principal components analysis by social scientists. Economists, sociologists, and especially psychologists believe that correlations between transformed orthogonal variables are a revelation of the 'real' structure of the world. Biologists are apparently unaware that in constructing the correlation analysis itself they impose a model on the world. Their assumption is that they are approaching the data in a theory-free manner and that data will 'speak to them' through the correlation analysis" (Lewontin, 1974, 156).

If these common practices with correlational analyses are viewed from the vantage point of a mapping, or two-place account of representation, then the relation between the formal systems and what structure they have taken on from empirical data is understood as a brute fact. What story such correlational analyses tell is not subject to refinement or negotiation; the methods simply reveal structure in the world that they share with the world. If, on the other hand, the inference-licensing representation between model and target is taken to exist only in virtue of a human interpreter, our analysis shifts drastically. With such a view of representation in mind, statistical analysis is then understood as a process which rests essentially on human input and decision-making. A researcher has within their power the capacity to use such statistical methods in an open-ended number of arbitrary ways, drawing erroneous conclusions from erroneous mapping of models to targets.

Lewontin's criticism of the use of ANOVA, linear and multiple regression, and PCA in data-intensive sciences rested on a particular philosophical conception of the epistemic function of formal analysis and the role of human choices therein. Critically, on Lewontin's analysis, it is ultimately an interpretive act on the part of a human cognitive agent which lends the model any meaning at all, which grants it ultimately its epistemic purchase over nature. Without recognizing the role of the interpreter in this exercise, the methodological error remains undetectable, and the procedure is granted undo legitimacy.

4.5 Conclusion

In the course of this chapter, I have attempted to show that agent-relative or triadic accounts of scientific representation have an as yet unrecognized advantage over mapping or dyadic accounts, because they alone equip us to diagnose and correct reification. I hope to have established: 1. That errors of reification are relatively common in scientific modeling practice. 2. That errors of reification come in many forms, of which I have offered something of a loose taxonomy. 3. That fallacies or reification are an issue particular to scientific representation. 4. That accounts of scientific representation which take representation to be a dyadic relation, often called "mapping accounts," obscure reification. 5. That agent-relative accounts of representation, by contrast, make it easier to detect and to correct for reification. With the exploration of case studies in Section 3, I hope to have demonstrated how variant representational accounts alter how we assess reification. While most philosophers of science in the representation debate who initially endorsed mapping views now endorse agent relative views, this ability to handle reification should count as an additional point in favor of triadic accounts.

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